

VIET NAM NATIONAL UNIVERSITY HO CHI MINH CITY
UNIVERSITY OF ECONOMICS AND LAW

FACULTY OF INFORMATION SYSTEMS



FINAL PROJECT REPORT

AN MOBILE-COMMERCE APPLICATION FOR
EYE-GLASS COLLAB BRAND

Supervisor: Tran Duy Thanh, Ph.D.

Group: COLLAB

Class: 243MI2703

Ho Chi Minh City, June/2025

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Subject: Mobile-Commerce
Development

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Group: Group COLLAB

No.	Member's Full Name	Student ID	Position
1	Lưu Hà Vy	K224111509	Leader
2	Nguyễn Thị Bảo Ngân	K224111493	Member
3	Nguyễn Ngọc Bảo Tiên	K224111505	Member
4	Nguyễn Thị Tường Vy	K224111509	Member
5	Mai Thị Thu Quyên	K214110847	Member

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Once again, the group would like to sincerely thank the teacher for his companionship and inspiration throughout the past semester.

Our team sincerely thank you!

COMMITMENT

We, the members of Group COLLAB, solemnly affirm that this final report for the course “*Mobile-Commerce Development*” has been completed with honesty, diligence, and in full adherence to academic standards.

All findings, discussions, and conclusions contained in this report reflect our joint efforts and original work, except where external sources have been clearly acknowledged through proper citations and references.

We are grateful for the guidance and support of PhD. Tran Duy Thanh, whose supervision has played an essential role in shaping the direction and quality of our project.

We fully recognize the importance of academic integrity and understand that any breach of ethical conduct - such as plagiarism or fabrication - is unacceptable and subject to institutional penalties. With this declaration, we take full responsibility for the authenticity, accuracy, and integrity of our submitted work.

Ho Chi Minh City, June 2025

Signed by COLLAB team

TABLE OF CONTENTS

ACKNOWLEDGEMENT	i
COMMITMENT.....	ii
LIST OF ABBREVIATIONS.....	xi
ABSTRACT.....	xiii
CHAPTER I. PROJECT ANALYSIS	1
1.1. Overview of the project.....	1
1.1.1. Reason for choosing the topic	1
1.1.2. The users that the system serves	2
1.1.3. Competitor research.....	3
1.1.4. Goals.....	4
1.1.5. Business overview.....	5
1.1.6. Products.....	7
1.1.7. App requirements	8
1.2. Theoretical fundamentals	11
1.2.1. Android Studio.....	11
1.2.2. Firebase	11
1.2.3. Angular	12
1.2.4. Github	12
1.2.5. AI Agent – n8n	13
CHAPTER II. SYSTEM ANALYSIS AND DESIGN.....	14
2.1. Actors and use cases.....	14
2.2. Business process modeling notation (BPMN).....	18
2.2.1. Login and registration	18

2.2.2.	<i>Cart and order</i>	18
2.2.3.	<i>Change user information</i>	19
2.2.4.	<i>Dynamic promotion</i>	20
2.2.5.	<i>Admin</i>	20
2.3.	Database	21
2.3.1.	<i>Define collections of database</i>	21
2.3.2.	<i>Entity Relationship Diagram (ERD)</i>	30
2.3.3.	<i>Data Flow Diagram (DFD)</i>	30
2.4.	Sitemap	35
2.5.	Mockup	36
2.5.1.	<i>Logo and favicon design</i>	36
2.5.2.	<i>Client side</i>	37
2.5.3.	<i>Admin side</i>	43
2.6.	n8n AI agent	45
CHAPTER III. DEVELOPMENT APPLICATION		47
3.1.	Create sample data for Firebase Realtime	47
3.2.	Develop system modules	47
3.2.1.	<i>Client side</i>	47
3.2.2.	<i>Admin side</i>	61
CHAPTER IV. CONCLUSION		68
REFERENCES		70
APPENDIX A – DOCUMENT REFERENCES		a
APPENDIX B – GITHUB LINK FOR MOBILE APP		a
APPENDIX C – GITHUB LINK FOR ADMIN WEB		a
APPENDIX D – WEBSITE MOCKUP ON FIGMA		a
APPENDIX E – FIREBASE REALTIME DATABASE		a

APPENDIX F – SCIENCE PAPER	a
APPENDIX G – EVINDENCE CLIP	a
APPENDIX H – PRESENTATION FILE	b
APPENDIX I – DIAGRAMS	b
APPENDIX K – EXCEL WORKING CONTRIBUTION	b

LIST OF FIGURES

Figure 1. 1: Collab's business model canvas.....	7
Figure 1. 2: Frame glasses.	7
Figure 1. 3: Sunglasses.	8
Figure 2. 1: User's Use Case.	14
Figure 2. 2: Administrator's Use Case.	16
Figure 2. 3: Login and registration BPMN.	18
Figure 2. 4: Order and cart BPMN.	18
Figure 2. 5: Profile update BPMN.	19
Figure 2. 6: Dynamic promotion BPMN.	20
Figure 2. 7: Admin Management BPMN.	20
Figure 2. 8: E-R Diagram.	30
Figure 2. 9: Login and Registration DFD.	30
Figure 2. 10: Cart and order DFD.	31
Figure 2. 11: Change user information DFD.	32
Figure 2. 12: Dynamic Promo DFD.	33
Figure 2. 13: Admin DFD.	34
Figure 2. 14: Sitemap.	35
Figure 2. 15: Image of Collab Logo	36
Figure 2. 16: Image of Collab Favicon	37
Figure 2. 17: Color palette.	37
Figure 2. 18: Home Page (Login & None Login).	37
Figure 2. 19: Notification.	38
Figure 2. 20: About Us & Policy.	38
Figure 2. 21: Details Page and Component.	38
Figure 2. 22: Customer Reviews.	39
Figure 2. 23: Customer profile & Edit Information.	39
Figure 2. 24: Edit Profile picture.	40

Figure 2. 25: Customer choose to take picture on the phone.	40
Figure 2. 26: History.....	40
Figure 2. 27: Login.	41
Figure 2. 28: Sign-up new Account.....	41
Figure 2. 29: Missing variable example.	42
Figure 2. 30: Cart and Sum-up Pop-up.....	42
Figure 2. 31: Cart processing.....	42
Figure 2. 32: Check out process.	43
Figure 2. 33: Admin login design.	43
Figure 2. 34: Dashboard.....	43
Figure 2. 35: Account Administration.....	44
Figure 2. 36: Promotion Administration.....	44
Figure 2. 37: Product Administration.	44
Figure 2. 38: Order Management.....	45
Figure 2. 39: n8n dynamic promotion workflow.....	45
Figure 3. 1: Login page.....	47
Figure 3. 2: Forgot password.	48
Figure 3. 3: Register account.	48
Figure 3. 4: Create account.	49
Figure 3. 5: Popup confirms successful account registration.	49
Figure 3. 6: Welcome page.	50
Figure 3. 7: Pop-up login.	50
Figure 3. 8: Search bar.....	51
Figure 3. 9: Product category.....	51
Figure 3. 10: Filter & Sorting.	52
Figure 3. 11: Product details.	53
Figure 3. 12: Profile page.	53
Figure 3. 13: Edit profile.....	54
Figure 3. 14: Change avatar.....	54

Figure 3. 15: Notification page.....	55
Figure 3. 16: Notification details.	55
Figure 3. 17: Shopping cart page.....	56
Figure 3. 18: List of vouchers.....	56
Figure 3. 19: Checking total cost, applying voucher before checkout.	57
Figure 3. 20: Checkout.....	57
Figure 3. 21: User chooses Visa/MasterCard payment method.	58
Figure 3. 22: Edit personal information shipping.....	58
Figure 3. 23: Pop-up confirm checkout.	59
Figure 3. 24: Order successfully page.	59
Figure 3. 25: Order history.	60
Figure 3. 26: Review product page.....	60
Figure 3. 27: Policy page.	61
Figure 3. 28: Admin login page.....	61
Figure 3. 29: Admin forgot password page.	62
Figure 3. 30: Admin confirm code page.....	62
Figure 3. 31: Admin reset password page.	62
Figure 3. 32: Admin homepage with dashboard.....	63
Figure 3. 33: Product list.....	63
Figure 3. 34: Editing product.....	64
Figure 3. 35: Add new product.	64
Figure 3. 36: Delete product.	64
Figure 3. 37: Promotion list.	65
Figure 3. 38: Editing promotion information.	65
Figure 3. 39: Add a new promotion.....	65
Figure 3. 40: Delete promotion.....	66
Figure 3. 41: Order list.....	66
Figure 3. 42: Filter order based on status	66
Figure 3. 43: Admin account list.	67

Figure 3. 44: Add a new admin.	67
Figure 3. 45: Edit info admin.....	67
Figure 3. 46: Delete admin account.	67

LIST OF TABLES

Table 1. 1: The users that the system serves.....	2
Table 1. 2: Competitor analysis.	3
Table 1. 3: Collab introduction.	5
Table 2. 1: Application's functions to User.....	15
Table 2. 2: Application's functions to Administrator.	17
Table 2. 3: Structural differences between SQL and NoSQL.	21
Table 2. 4: Collections of database.....	22

LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
APIs	Application Programming Interfaces
B2C	Business-to-Consumer
BPMN	Business Process Modeling Notation
CLI	Command Line Interface
CRUD	Create, Read, Update, Delete
DFD	Data Flow Diagram
E-R diagrams	Entity-Relationship Diagrams
FCM	Firebase Cloud Messaging
ID	Identifier
IDE	Integrated Development Environment
ISO	International Organization for Standardization
JSON-based	JavaScript Object Notation-based
LLM	Large Language Model
n8n	Node-based Workflow Automation Tool (<i>“n8n” = “nodemation”</i>)
NoSQL	Not Only SQL (Non-relational Database)

OTP	One-Time Password
SQL	Structured Query Language
SPAs	Single-page Web Applications
UI	User Interface
UX	User Experience
UV	Ultraviolet

ABSTRACT

This report presents the process of analyzing, designing and developing the Collab Eyewear e-commerce application, which specializes in providing fashion eyewear products such as frames and sunglasses. The system includes a mobile application for users and an administrative website for administrators. The application is built on the Android Studio platform, using Firebase Realtime Database to store and synchronize data in real time, and integrating AI Agent technology (n8n) to personalize promotions based on shopping behavior.

The project not only meets basic functions such as login, shopping cart, payment, and order management, but also enhances the user experience with a modern interface, supporting smart filtering and searching. On the administrator side, the system allows for intuitive management of products, categories, promotions, orders, and revenue statistics. Through the implementation process, the team has applied a lot of knowledge about e-commerce application development, while improving teamwork skills, system design and project process management.

CHAPTER I. PROJECT ANALYSIS

This chapter outlines the foundation for the Collab project by defining its key requirements and theoretical background. It begins by presenting the rationale behind choosing the topic and identifying the primary users that the system aims to serve. Through competitor research and market observation, the chapter highlights current trends and potential gaps that Collab is designed to address. The project's goals, business model, and product offerings are also clarified to ensure a clear direction for development.

Furthermore, this section establishes the technical and functional requirements of the application to align with user expectations and business objectives. Finally, the chapter introduces the theoretical fundamentals and tools applied throughout the project, serving as the academic and technical backbone for system design and implementation.

1.1. Overview of the project

1.1.1. Reason for choosing the topic

Nowadays eyeglasses have evolved from being purely vision-correction tools into fashion accessories that reflect one's personality and style. However, many consumers still struggle to find frames that suit their face shape and personal preferences, often making the shopping experience inconvenient and less enjoyable.

Nearly 70% of users feel confusion or nuisance when purchasing glasses online due to lack of personalized recommendations resulting in inefficient and time-consuming shopping (Statista, 2024). Meanwhile, current platforms mainly focus on product diversification without paying attention to the feature of consulting and recommending glasses based on each user's unique facial features and aesthetic taste. For this reason, this project will be chosen to develop “*Collab*” as an Android mobile application commerce project. This application not only provides high-quality eyewear products but also integrates a smart recommendation feature that helps users easily discover the most suitable frames.

Furthermore, with the rapid growth of mobile application shopping and the shift in consumer behavior towards on-the-go purchases, there is a clear need for a modern, user-friendly mobile application solution. With Collab, we aim to deliver not only

quality products but also an enjoyable, convenient, and personalized mobile application shopping experience that builds confidence and satisfaction for every user.

1.1.2. The users that the system serves

Table 1. 1: The users that the system serves.

No.	Personal information	Description	Key needs & Expectations
1	Young professionals (Aged 25 - 34)	Highly engaged in fashion trends and awareness of eye health, actively follows lifestyle influencers, and has disposable income to spend on quality products.	- Stylish eyewear as both a fashion accessory and protective tool (e.g., UV-blocking, blue light-filtering). - AI recommendations. - Easy online purchase.
2	Gender	Both men and women are potential customers, but women tend to invest more in fashion products, making them a key focus group.	- Trendy eyewear designs for women and men - Focus on offering stylish yet functional eyewear.
3	Middle-income customers	Individuals who are willing to invest in quality products but are also price-sensitive, preferring affordable options.	- Transparent pricing - Value-for-money offers - Promotions & discounts
4	Occupation	Includes students, early-career professionals, and those working in smart or casual style fields. They value both style and practicality.	- Practical and stylish eyewear for both work and social settings - Comfortable, durable options

5	Geography	Urban areas, particularly in cities with a growing middle class such as Hanoi, Ho Chi Minh City, and Da Nang.	- Prefer online shopping, convenient - Friendly application interface, easy to use - Fast delivery service, good online support.
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1.1.3. Competitor research

In Vietnam, the eyewear market is highly fragmented, with small businesses accounting for approximately 82% of the market share (Statista, 2024). While major international brands like Stella McCartney, Premio, Air Optix, and Crizal hold smaller portions, they remain competitive by offering high-end products.

The primary direct competitors in the affordable segment - targeting the same customer base - include HMK Eyewear, Lily Eyewear, and Anna Eyewear. The following table provides a detailed comparison of these competitors.

Table 1. 2: Competitor analysis.

	HMK	LiLy	Anna
Strength	- Offers a wide range of eyewear products with transparent sourcing. - Competitive pricing suitable for various customer segments. - Great after-sales care and warranty policies.	- Established in 2018, Lily Eyewear is popular for young consumers as it offers a variety of fashionable styles. - Reasonable prices for young consumers. - Wide range of products and attentive customer service with an enthusiastic experienced team.	- 13+ years in development staying on top of trends and innovating. - Excellent shopping experience and high quality, reliable products. - 50+ branches within Vietnam.

			- Reasonably priced products.
Weakness	- Limited variety in eyewear models from globally recognized fashion brands. - Does not support diverse payment methods (Cash on delivery and bank transfer via QR code only).	- Limited variety in eyewear models from globally recognized fashion brands. - No payment options through ZaloPay or MoMo, which limits convenience for customers (Cash on delivery and card payment only).	- Limited variety in eyewear models from globally recognized fashion brands. - Cash on delivery is the only payment method that is available.

1.1.4. Goals

The primary goal of the project is to develop an e-commerce Android mobile application for Collab eyewear, with concurrent focus on the User Interface (UI) and User Experience (UX) and also delivering a personalized experience. The application's interface is designed to apply across all devices with attention to detail such as engaging design elements and a lighter color palette that reflects the brand, leading to increased customer engagement and comfort.

Functionality is vital, as the app is developed to allow customers to browse, select, order, track, and leave reviews after purchase. The system also features AI-driven dynamic promotions that generate personalized offers based on their purchase history, as well as include pushing notifications to keep customers informed in real-time. Meanwhile, administrators are empowered with tools to manage product listings,

promotions, process orders, and access sales tracking analytics. Designed for growth, it adapts to customer needs and industry trends for long-term success.

1.1.5. Business overview

a. Collab introduction

Table 1. 3: Collab introduction.

No.	Feature	Description
1	Brand Name	Collab
2	Project Idea	Collab is an innovative online eyeglasses brand, offering stylish, comfortable, and affordable glasses through a seamless digital shopping experience.
3	Slogan	Your Vision, Our Collaboration
4	Inspiration	Collab was founded on the belief that great things are achieved when people work together. The brand name “<i>Collab</i>” represents collaboration, unity, and partnership - a shared effort to bring clarity, style, and innovation to the eyeglasses industry. Our slogan, “<i>Your Vision, Our Collaboration,</i>” reflects our commitment to working closely with customers - offering stylish frames, personalized solutions, and a seamless shopping experience. More than just a necessity, eyeglasses at Collab blend fashion, function, and individuality, ensuring everyone finds the perfect pair to match their vision and style.
5	Project Type	Business-to-Consumer (B2C)

6	Vision	Within the next five years, Collab aims to be Vietnam's leading and most trusted online eyewear brand, known for quality, convenience, and reliability. Our goal is to revolutionize digital eyewear shopping, expand regionally, and ultimately become a globally recognized brand.
7	Mission	To empower individuals with eyeglasses that enhance their confidence, vision, and individuality through a convenient and innovative online platform.
8	Core Values	- Continuously improving our mobile platform to provide a seamless and engaging shopping experience. - Leveraging advanced technology. - Ensuring premium craftsmanship and materials in every product. - Offering a diverse range of styles and sizes for everyone, tailored to individual preferences. - Committing to eco-friendly practices to protect the planet. - Prioritize fast delivery, secure payments, and excellent customer support.

b. Business goals

- Short-term goal: To provide a specialized and refined online space where customers - especially our target audience - can freely explore the world of eyewear with a superior personalized experience. The Android mobile application prioritizes convenience, enabling users to easily access products and services.
- Long-term goal: Collab aims to be a leading and trusted eyewear brand, setting the standard for online shopping and personalized consultation. We strive to capture 5% of the market by offering diverse, high-quality eyewear and tailored advisory services. Beyond business, we are committed to social impact and sustainability through community support, eco-friendly materials, and waste reduction, building lasting trust and enhancing both eye care and personal style.

c. Business model canvas

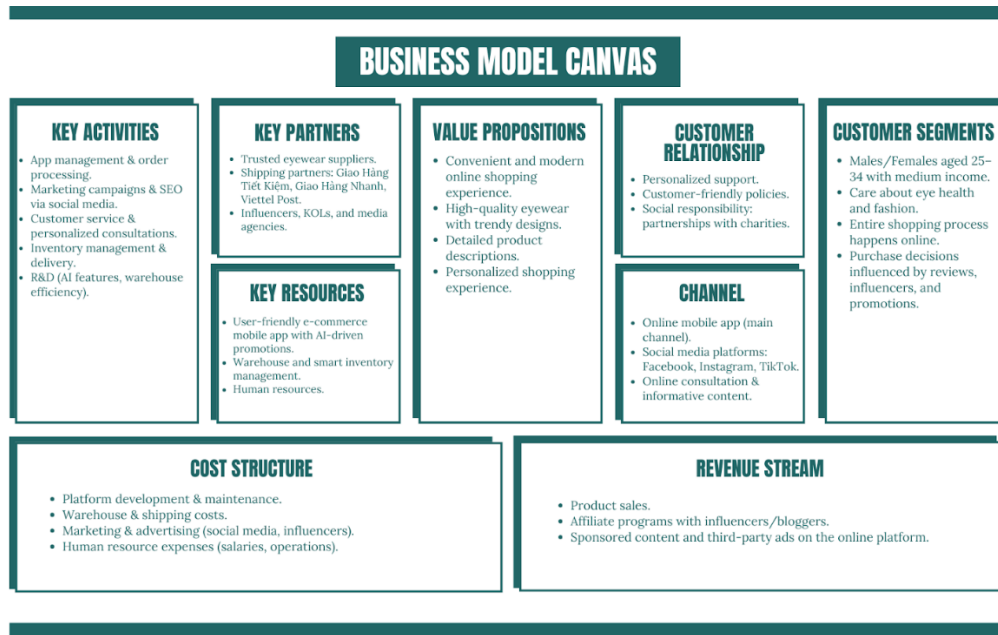


Figure 1. 1: Collab's business model canvas.

1.1.6. Products

Our mobile application offers a wide selection of frame glasses and sunglasses, designed to meet the diverse needs and preferences of customers. The frames and sunglasses we offer showcase both classic and contemporary styles, and each product is presented with high-quality images, detailed descriptions, specifications, and transparent pricing. Specifically, our mobile platform allows users to browse, compare, and select products through a quick and easy interface that makes the shopping experience simple and easy. In addition, items in our catalog are also regularly updated to reflect new seasonal trends, allowing customers to maintain their elegant, stylish confidence each season.

Some sample of Collab's glasses:



Figure 1. 2: Frame glasses.

With frame glasses that are lightweight, durable, and available in various styles to suit every customer's taste, customers can choose from classic, trendy, or professional

designs. Each frame is made from high-quality materials to ensure comfort and long-lasting wear.



Figure 1. 3: Sunglasses.

Our sunglasses provide excellent UV protection while keeping customers stylish. They are offered in multiple shapes and colors to match different face shapes and fashion preferences. Designed for comfort, they are perfect for everyday outdoor use.

1.1.7. App requirements

This mobile application is built to support the online sale of eyeglass frames and sunglasses. It aims to provide users with a seamless, intuitive shopping experience similar to modern e-commerce platforms. The app also integrates AI technologies and Firebase for enhanced personalization and security.

a. Functional requirements

Regarding to Client functional requirements:

- Product catalog: The application includes a comprehensive product catalog that allows users to browse, filter, and search for eyeglasses and sunglasses by various criteria such as type, material, color and price range. Moreover, the app supports either infinite scrolling or pagination, ensuring smooth performance and easy navigation as users browse through numerous items.
- Product details: In this screen, the app displays detailed information including specifications (frame material, lens type, dimensions, etc.), descriptions, and user reviews along with rating.
- Product review: Users can rate and write comments about the product that they had already bought, and can also see other users' reviews and comments about the product.
- User authentication (Firebase): Authentication is handled via Firebase, ensuring secure and scalable user login and registration.

- User profile management: App users are allowed to update all previously registered personal information, including their name, email, phone number, display name, and avatar. Additionally, users are permitted to add multiple shipping addresses.
- Shopping cart: Enables logged-in users to manage their selected products with ease. Users can add or remove items from the cart as they browse the catalog, and adjust the quantity of each product directly within the cart interface. This ensures a seamless and personalized shopping experience tailored to individual user preferences.
- Checkout process: To streamline purchases by guiding users through a clear and efficient sequence. It supports multiple payment options, allowing customers to choose their preferred method. During checkout, users can select from existing shipping addresses or input a new one, ensuring flexibility in delivery preferences. The process also shows a comprehensive order summary, including itemized costs and the total amount due. Additionally, users can choose their desired shipping method, each accompanied by estimated delivery times.
- Order history: Customers can access their order history in order to track all of their orders, which includes details such as order ID, product names, order status, and date of purchase.
- AI-Driven features: An unique and important feature of this application which adapts to individual user behavior. The app delivers dynamic promotions tailored to user activity, such as viewing patterns or past purchases, ensuring that marketing content remains relevant and compelling.
- Notification system: Designed to keep users informed and engaged throughout their shopping journey. Users receive real-time updates on order status, including confirmations, shipping progress, and delivery notifications. Promotional alerts are also sent to highlight discount offers, special deals, and especially personalized campaigns. Furthermore, with the aim of encouraging purchase completion, the app sends reminders for abandoned carts, prompting users to return and finalize their orders.

Regarding to Administrator functional requirements: provides a comprehensive backend interface for managing the system's core data and monitoring business

performance. Authentication for administrators is handled through Firebase Authentication, ensuring secure login and role-based access control.

- Admin can perform full CRUD (Create, Read, Update, Delete) operations on product and promotional datasets stored in Firebase. They can add new eyewear items or more new categories, as well as update product information, apply or remove promotional offers, and delete outdated entries.
- Order management is a critical function, allowing administrators to view and update the status of customer orders. This includes tracking order progress, confirming shipments, processing cancellations, and handling fulfillment issues. Additionally, the admin interface provides access to detailed customer profiles, including interaction histories, purchase records, and engagement statistics, which can be used to improve customer service and tailor personalized promotional campaigns.
- The website includes a sales tracking feature that displays sales data through visual diagrams. These diagrams help administrators identify sales trends, assess the effectiveness of marketing campaigns, and make data-driven decisions to optimize operations and revenue.

b. Non-functional requirements

- Performance requirements
 - The application must ensure a response time of under 2 seconds for common actions such as product search, adding to cart, and displaying order details.
 - The admin page needs to efficiently handle tasks involving large data management, such as order review and revenue statistics, with a maximum latency of no more than 3 seconds.
- Security
 - Both the client application and the admin website must use Firebase Authentication to ensure secure login and account management.
 - User and order data must be encrypted during transmission and storage.
 - The system must have a clear role-based access policy between users and administrators, preventing unauthorized access to internal data.
- Usability

- The user interface of the application must be user-friendly and easy to navigate. All key functions such as product filtering, placing orders, and tracking orders must be easy to find and operate.
- The admin page must also feature an intuitive layout, enabling administrators to work efficiently with charts, data tables, and filtering tools.

1.2. Theoretical fundamentals

1.2.1. Android Studio

Android Studio is the official integrated development environment (IDE) provided by Google for building Android applications. It is based on the IntelliJ IDEA platform and integrates all the necessary tools, from coding and designing the interface to compiling and testing. (TechTarget, n.d.). Android Studio offers several notable benefits for the Android app development process. First, it strongly supports two popular programming languages, Java and Kotlin, allowing developers flexibility in choosing the appropriate technology. Additionally, according to Android Developers, the drag-and-drop design interface enables users to quickly build the user interface (UI) without having to write extensive code. Android Studio also comes with Gradle, a powerful tool for managing versions, libraries, and supporting an efficient project build process. Notably, the Android Emulator allows developers to test applications directly without the need for physical devices, saving time and improving work efficiency. In this project, Android Studio is used to develop the main code of the mobile application, including handling the user interface, functionalities such as placing orders, filtering products, displaying glasses frames, and connecting data with Firebase.

1.2.2. Firebase

According to Google (n.d.), Firebase is a development platform from Google that provides cloud services such as databases, user authentication, file storage, and app analytics. The Firebase platform allows real-time data synchronization when changes occur from the admin website and the end-user application. Firebase Authentication provides secure storage and simplifies the process for developers. Firebase Storage helps in storing images, such as product images or user avatars, efficiently. Leveraging these advantages, the project will use Firebase to store and manage data such as user accounts,

orders, products, and frame images, along with user activity data that supports personalized promotional code features. Furthermore, the login and registration authentication function is implemented through Firebase Authentication.

1.2.3. Angular

Angular is an open-source web application framework developed by Google, designed for building dynamic, single-page web applications (SPAs). It is based on TypeScript and provides a comprehensive development toolkit, including features such as two-way data binding, dependency injection, and a modular, component-based architecture. These features help streamline development and make code easier to maintain and scale (GeeksforGeeks, 2024).

Angular supports efficient navigation through built-in routing, and its integration with RxJS enables reactive programming for responsive data handling. The Angular CLI (Command Line Interface) offers powerful tools for generating components, testing, and deploying applications effectively.

In this project, Angular is used to develop the admin web application, allowing administrators to manage products, orders, and promotions. Angular is tightly integrated with Firebase, enabling real-time data synchronization, user authentication, and efficient backend communication without full-page reloads. Its structured framework and strong tooling contribute significantly to the speed and reliability of the web-based admin system.

1.2.4. Github

GitHub is an online platform for source code storage and management using the Git version control system, allowing multiple developers to work together on a project with features for tracking changes and merging code. GitHub provides a storage space that facilitates collaboration among team members, helping programmers track the history of code edits clearly. Moreover, it improves the software development process by automating build, test, and deployment processes, enabling code changes to be transferred more efficiently and securely. GitHub is used to store the project's source code, support teamwork, track progress, and synchronize code versions during the development of the application (Brown, 2016).

1.2.5. AI Agent – n8n

n8n is an open-source workflow automation tool that allows for connecting services and automating tasks. When combined with artificial intelligence (AI agent), n8n can process data and suggest intelligent actions based on context. n8n helps automate the process of sending personalized promotions to users, enhancing the personalized experience. n8n is relatively easy to integrate with services like Firebase, Gmail, Telegram, and internal APIs. With its intuitive drag-and-drop interface and minimal coding requirements, n8n is suitable for both professionals and non-experts. In this project, the AI agent is used in combination with n8n to suggest personalized promotional campaigns based on users' order history or behavior.

CHAPTER II. SYSTEM ANALYSIS AND DESIGN

This chapter presents the system analysis and design process for the Collab application, including identifying actors – use cases, modeling business processes using BPMN, designing databases with E-R diagrams and DFDs for each main function. In addition, sitemaps and interface mockups for both the user and admin sides are also built to ensure intuitiveness and efficiency of use. Finally, integrating n8n AI agents helps support automation and enhance user experience.

2.1. Actors and use cases



Figure 2. 1: User's Use Case.

The Client's Use Case diagram provides a high-level overview of how the user (customer) interacts with the e-commerce system. This diagram captures the main and supplementary functions, scenarios of many of the functions are contingent on users being logged in to enable secure and personalized staircase flows. The following table lists every user function with a description, and whether or not being logged in is a prerequisite to performing the action.

Table 2. 1: Application's functions to User.

Function	Login Required	Description
View Terms, Policies, Conditions, About Collab	-	Allows customers to view platform policies, terms of use, and information about the Collab brand.
Login/Logout	-	Manages user login and logout through Firebase Authentication to enable secure access to the platform.
View Product Catalog	-	Display a list of products that can be browsed, filtered, and searched by customers.
View Product Details	-	When a customer taps on a product, it shows detailed information (e.g., description, specifications, reviews).
Manage User Profile	x	Allows customers to view and update their profile information (e.g., name, email, gender).
Track Orders	x	Allows users to view the status of their current orders (e.g., "Pending", "Shipping", "Delivered") and access past order history with full details.

Give Feedback	x	Enables users to leave product reviews or ratings after an order has been successfully completed.
Manage Shopping Cart	x	Allows users to add or remove products from their cart in preparation for checkout.
Proceed to Checkout	x	Facilitates the purchase of items in the cart, including order review, payment, shipping details, and optional discount usage.
Receive AI-Driven Promotions	x	Displays personalized, AI-generated promotions based on customer purchase history.
Receive Push Notifications	x	Sends personalized notifications to customers about promotions, order updates, and other important alerts.

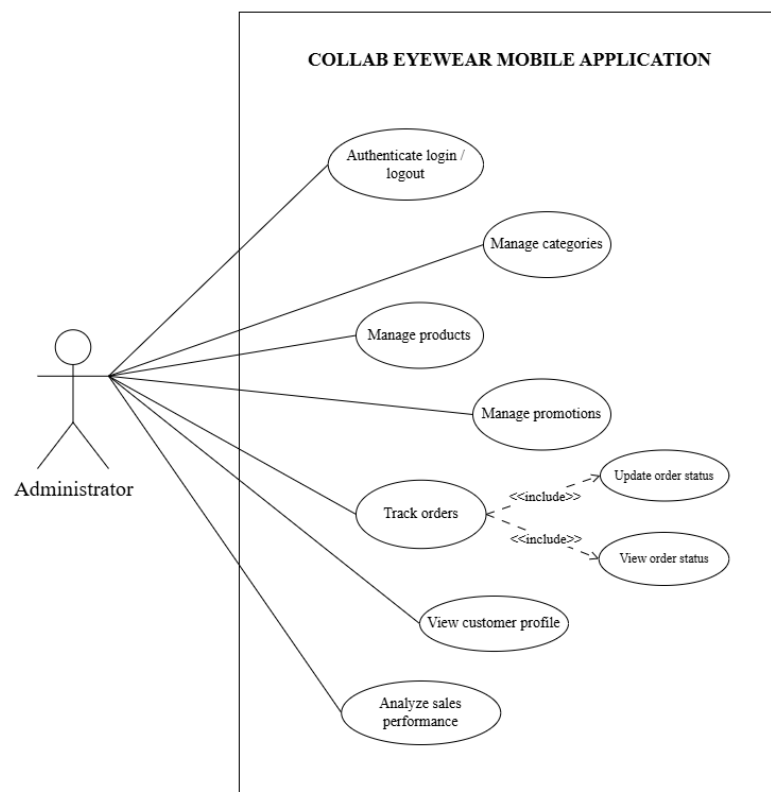


Figure 2. 2: Administrator's Use Case.

The Administrator's Use Case diagram displays the administrative control of the back-end processes of the e-commerce system. To perform any CRUD operations in their management sections, admins must be securely logged in. The table summarized below includes the primary tasks of an administrator, appropriate descriptions of each process.

Table 2. 2: Application's functions to Administrator.

Function	Description
Login / Logout Authentication	Manages admin login and registration through Firebase for secure access.
Manage Product	Provides full CRUD operations on the product dataset stored in Firebase.
Manage Categories	Enables the creation, modification, and deletion of product categories (e.g., frame eyeglasses, sunglasses).
Manage Promotions	Provides full CRUD operations on the promotion dataset stored in Firebase.
Track Orders	Allows admins to view and update the status of customer orders, including processing, shipment, and cancellations.
View Customer Profile	Gives admins access to user profiles, including purchase history, engagement patterns, and activity logs for monitoring purposes.
Analyze Sales Performance	Displays a visual representation (diagram) of sales data for analysis and decision-making.

2.2. Business process modeling notation (BPMN)

2.2.1. Login and registration

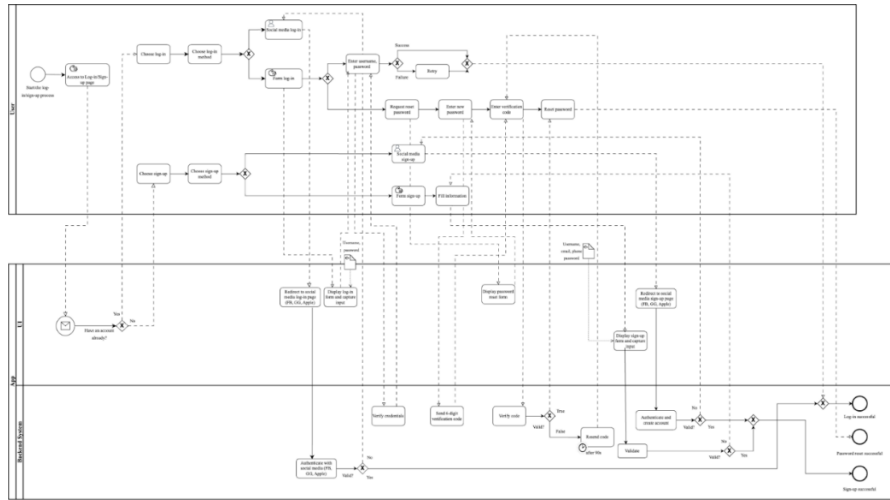


Figure 2. 3: Login and registration BPMN.

This BPMN diagram presents the user authentication workflow of the Android mobile application, encompassing both the login and sign-up functionalities. The process supports multiple authentication methods, allowing users to either log in or register via form-based input (using a username, email, phone number, and password) or through third-party social media platforms such as Facebook, Google, or Apple. In the event of a forgotten password, users can initiate a password recovery flow by receiving and entering a 4-digit verification code, followed by setting a new password upon successful verification. The flow incorporates key decision points (e.g., checking account existence, credential validity, and verification success) and leads to one of three main outcomes: login successful, sign-up successful, or password reset successful. The diagram is structured across four swimlanes - User, UI, App, and Backend System - to clearly delineate responsibilities and interactions among system components.

2.2.2. Cart and order

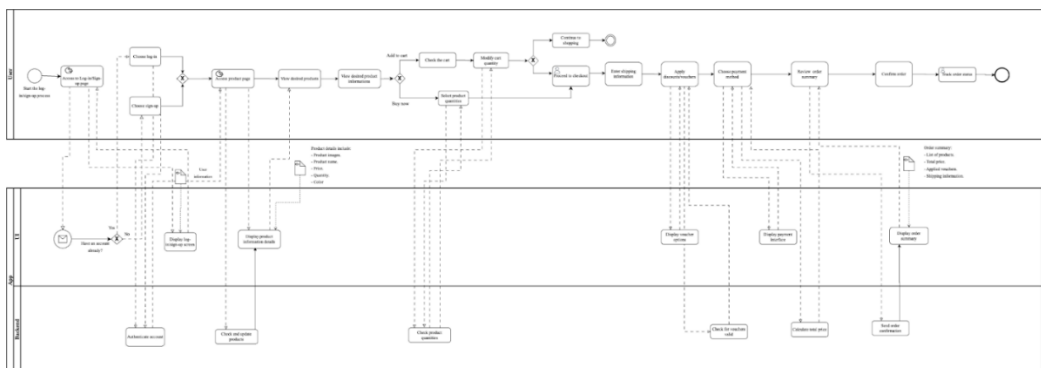


Figure 2. 4: Order and cart BPMN.

This BPMN diagram illustrates the mobile ordering and cart management workflow for an Android application. The process begins with users browsing products, viewing detailed information, and either adding items to their shopping cart or choosing to buy immediately. Users can check and modify cart quantities, then proceed to checkout, where they enter shipping information, apply vouchers or discounts, and choose a payment method.

The system then generates an order summary, including product details, applied vouchers, shipping data, and the total price, before users confirm their order and proceed to track order status.

The workflow is organized into four lanes:

- User: initiates and interacts with the shopping process.
- UI: handles user interface display and inputs.
- App: manages logic for product details, vouchers, and payment options.
- Backend: performs critical functions such as account authentication, inventory checks, voucher validation, price calculation, and order confirmation.

This end-to-end flow ensures a seamless shopping experience, with clear communication between the front-end and back-end systems.

2.2.3. Change user information

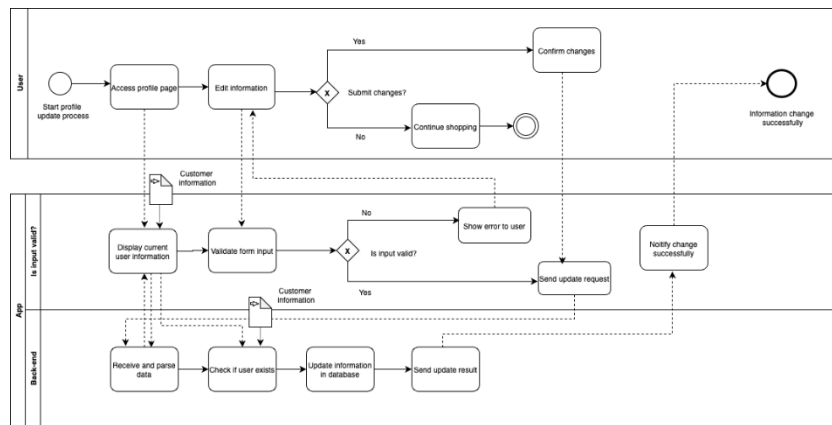


Figure 2. 5: Profile update BPMN.

This BPMN diagram illustrates the process of updating user profile information in a mobile application. The process begins when the user accesses the profile page and edits their information. If the user decides to submit the changes, the system validates the input. If the input is invalid, an error is shown; otherwise, a request is sent to the backend. The backend then parses the data, checks if the user exists, and updates the

information in the database. After successfully updating, a confirmation is sent back to the app, which notifies the user that the change was successful. If the user chooses not to submit the changes, they can continue shopping without any updates being made.

2.2.4. *Dynamic promotion*

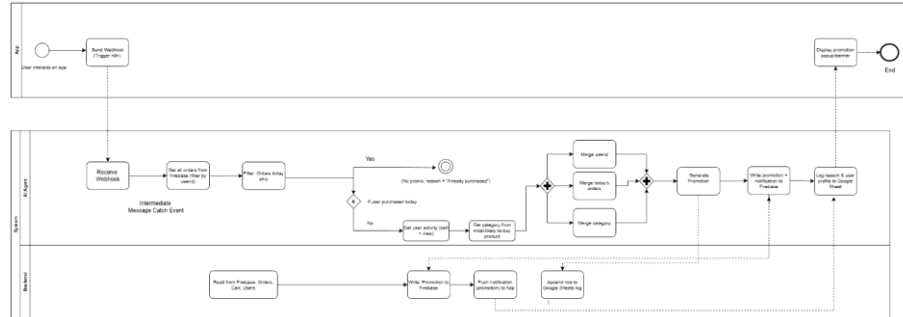


Figure 2. 6: Dynamic promotion BPMN.

The process starts when a user interacts with the app. The app sends a webhook (via n8n) to the system. The AI Agent receives the webhook and retrieves the user's order data from Firebase, filtered by user ID and today's date.

If the user has already made a purchase today → No promotion is generated → The reason is logged in Google Sheets.

If the user has not purchased → The system fetches the user's activity (e.g., cart actions, product views) → Identifies the category of the most-likely-to-buy product → Merges user ID, today's orders, and product category → Generates a personalized promotion.

The promotion is written to Firebase and a push notification is sent to the app. The user profile and promotion reason are logged to Google Sheets. Finally, the app displays a promotion popup/banner to the user.

2.2.5. *Admin*

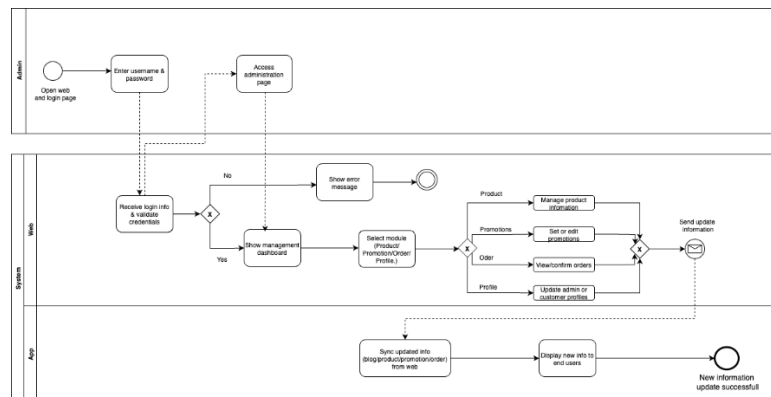


Figure 2. 7: Admin Management BPMN.

This BPMN diagram shows the process by which an Admin manages updates in a web-based administration system. The process starts when the admin opens the web and login page, then enters username and password. The system then validates the credentials. If valid, the admin is directed to the management dashboard; if not, an error message is displayed.

From the dashboard, the admin can select modules such as Product, Promotion, Order, or Profile. Each module allows the admin to perform different actions:

- Manage product information,
- Set or edit promotions,
- View or confirm orders,
- Update admin or customer profiles.

Once the admin completes an update, the system sends the updated information, which is then synchronized to the application. Finally, the app displays new info to end users, marking the update process as successful.

2.3. Database

2.3.1. Define collections of database

Database design is fundamental to the successful design of Collab Eyewear's system, as it directly impacts how data is stored, retrieved, and managed by the system in relation to many features, including user accounts, product management, orders, and analytics. A well-designed database will allow for an efficient, scalable, and reliable way for Collab to handle user interactions and internal business operations.

There are two main types of databases: SQL (relational) and NoSQL (non-relational). The difference between SQL and NoSQL structure is illustrated in table 2.3 below:

Table 2. 3: Structural differences between SQL and NoSQL.

SQL (Relational)	NoSQL (Non-relational)
Data stored in tables with fixed schemas	Data stored in flexible formats like documents or key-value pairs

Suitable for complex queries and transactions	Ideal for hierarchical data and real-time updates
Requires normalization	Encourages denormalization to reduce joins

This project uses Firebase Realtime Database, a NoSQL cloud-hosted (provided by Google) platform for storing data. It allows data to be stored as a JSON tree, particularly well-suited for this project due to its flexibility in handling nested and evolving data structures, such as user activity logs, shopping carts, and product reviews. Its ability to scale horizontally and provide real-time synchronization aligns perfectly with the dynamic and interactive nature of an e-commerce platform like Collab Eyewear. The table below outlines the main collections and their corresponding fields in the Firebase Realtime Database:

Table 2. 4: Collections of database.

No.	Collection Name	Field	Data Type	Description
1	Users	userid	string	Unique identifier for each user.
		name	string	User's full name.
		email	string	User's email address.
		username	string	User's login username.
		password	string	Encrypted password for authentication.
		gender	string	User's gender.
		phonenummer	string	User's phone number.
		defaultaddress	map	Default address. Includes keys: street, city, country.

		profilepicture	string	User's profile picture Base64.
		useractivity	array of maps	List of user activity records (embedded). Each map includes:
		└ activityid	string	Unique identifier for each activity record.
		└ action	string	The action taken by the user (e.g., “View Product”, “Added to Cart”).
		└ targetid	string	ID of the product involved.
		└ timestamp	string	Time when the activity occurred (ISO format).
2	Products	productid	string	Unique identifier for each product.
		categoryid	string	Reference to the category this product belongs to
		name	string	Name of the product.
		description	string	Product description.
		price	float64	Product price.
		stock	int64	Quantity in stock.
		image	string	Product image Base64.
		ratings	float64	Average rating score for the product, calculated from associated reviews.

3	Orders	orderid	string	Unique identifier for each order.
		userid	string	Reference to the user who placed the order.
		products	array of maps	List of products in the order (reference), each contains:
		└ productid	string	Reference to the product.
		└ quantity	int64	Number of items.
		└ price	float64	Price of the product.
		└ allowreview	boolean	A flag indicating whether the product can be reviewed (Only when status = "Delivered", all allowreview flags inside products will be set to true.).
		status	string	Current order status. - “Pending”: Order placed, awaiting processing. - “Shipping”: Order is on the way. - “Delivered”: Order has been delivered to customer → allowreview will be set to true.
		totalamount	float64	Total cost of the order before any discounts (sum of quantity × price for each product).

		promovalue	float64	The value of the promotion applied (e.g., 20000 for 20,000 VND).
		totalafterpromo	float64	Total amount after applying the promotion (Calculate by totalamount - promovalue).
		orderdate	string	Date the order was placed.
		shippingdetails	map	Object containing address (string), phonenumber (string), estimatedDelivery (timestamp).
		paymentmethod	string	Method of payment (“Cash on Delivery”, “Visa/ MasterCard”).
4	Reviews	reviewid	string	Unique identifier for each review.
		productid	string	Reference to the product being reviewed (ID of product from Products).
		userid	string	Reference to the user who left the review (ID from Users).
		rating	int64	Numerical rating (e.g., 1 to 5 stars).
		review	string	Textual feedback written by the user.
		orderid	string	Reference to the order in which the product was purchased (to

				ensure the review only comes after an order is complete).
		timestamp	string	Timestamp of when the review was created (ISO format).
5	Promotions	promotionid	string	Unique identifier for each promotion.
		userid	string/null	Reference to the specific user this promotion is assigned to. Set to null for public promotions.
		categoryid	string	Reference to the category that the promotion applies to.
		discounttype	string	Type of discount: “percentage” or “fixed” (e.g., “10%” or “50000 VND”).
		discountvalue	int64	Value of the discount (e.g., 10 for 10%, or 50000 for 50,000 VND).
		validfrom	string	Start date and time when the promotion becomes valid.
		validuntil	string	End date and time when the promotion expires.
		isused	boolean	Indicated whether the promotion code has been used (true) or not (false).
		promotioncode	string	Unique code used to apply the promotion.

6	Categories	categoryid	string	Unique identifier for the category.
		name	string	Category name (e.g., “Sunglasses”, “Frame glasses”).
		description	string	A short description of the category.
		imageurl	string	Image for the category.
7	Sales Data	salesid	string	Unique identifier for each sales record.
		date	string	Date of the sales record (daily granularity).
		totalsales	float64	Total sales generated on that specific day.
		productssold	array of maps	List of sold products (reference) on that day, each containing:
		└─ productid	string	Reference to the sold product.
		└─ productname	string	Name of the product.
		└─ quantity	int64	Quantity of that product sold.
		└─ revenue	float64	Revenue generated from that product (quantity × price).
8	Admin	adminid	string	Unique identifier for each admin.
		email	string	Admin’s email address.

		name	string	Admin's name.
		username	string	Admin's login username
		password	string	Encrypted password for authentication.
		lastlogin	string	Timestamp of the last login.
9	Cart	cartid	string	Unique identifier for each cart.
		userid	string	Reference to the user who owns the cart.
		products	array of maps	List of products in the cart (reference). Each object includes:
		└─ productid	string	Reference to the product.
		└─ quantity	int64	Number of items of the product in the cart.
		└─ price	float64	Price of the product at the time added to cart.
		totalamount	float64	Total cost of selected items in the cart (Calculated as sum of quantity x price for selected products).
		createdat	string	Timestamp of when the cart was created.
		updatedat	string	Timestamp of when the cart was last updated.

10	Notification	notificationid	string	Unique identifier for each notification.
		userid	string	Reference to the user this notification belongs to.
		type	string	Type of notification: - “Promotion” - Sent when there is a promotion assigned to the specific user. - “Order Status” - Sent when there is a change in the status of one of the user’s orders (e.g., “Pending”, “Shipping”, “Delivered”).
		title	string	Short title of the notification. The content of this field depends on the type.
		message	string	Detailed message of the notification. The content of this field depends on the type.
		relatedid	string	ID related to the subject: - If type = “Promotion”, this refers to the promotionid for the specific user that is assigned to. - If type = “Order Status”, this refers to the orderid.
		timestamp	string	When the notification was created.

2.3.2. Entity Relationship Diagram (ERD)

This E-R diagram depicts an e-commerce database system that captures information on users, products, orders, and related activities. It depicts major entities (Users, Products, Orders, Carts, Promotions, Reviews, User Activity, Categories, Sales Data and Admin) and their attributes and relationships. The design captures core capabilities, including order tracking, product organization, cart management, user activity management, and promotional discounting.

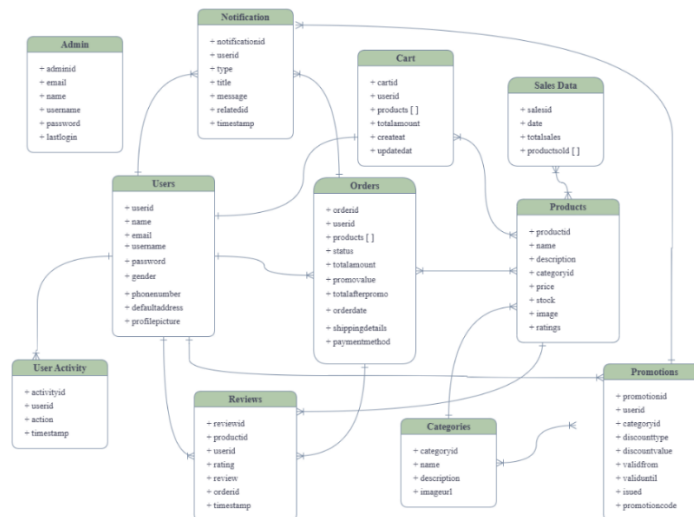


Figure 2. 8: E-R Diagram.

2.3.3. Data Flow Diagram (DFD)

a. Login and registration

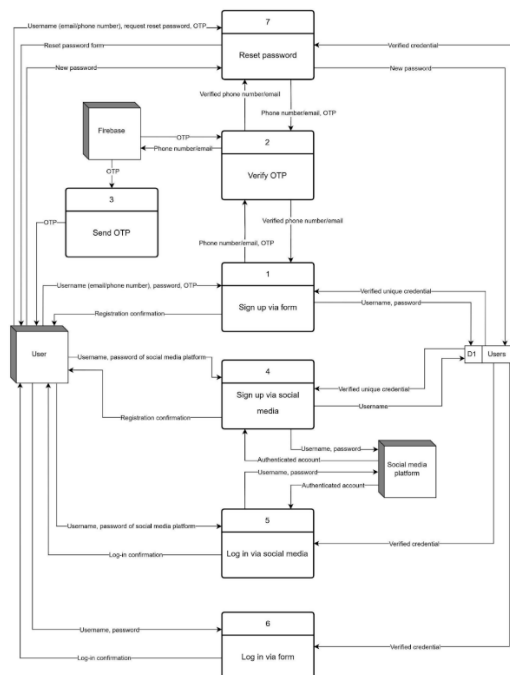


Figure 2. 9: Login and Registration DFD.

This Data Flow Diagram (DFD) illustrates the processes involved in user authentication, covering both registration and login functionalities. The system supports two registration methods: via a standard form and through social media platforms. For form-based registration, users input their username, email or phone number, password, and an OTP (One-Time Password). The system verifies the uniqueness of these credentials before storing the new account in the user database. OTPs are sent via Firebase and then validated to confirm the user's identity. Similarly, users can register using their social media credentials, which are verified through the respective third-party platform. Upon successful authentication, the account is either created or logged in.

For login, the process can also occur through social media or by entering a username and password into a form. In both cases, the credentials are validated, and if correct, the user gains access to the system. Additionally, a password reset feature is provided: users initiate this by submitting their username or email, receive an OTP via Firebase, and upon successful OTP verification, are allowed to create a new password. All verified credentials and user information are updated in the Users data store.

b. Cart and order

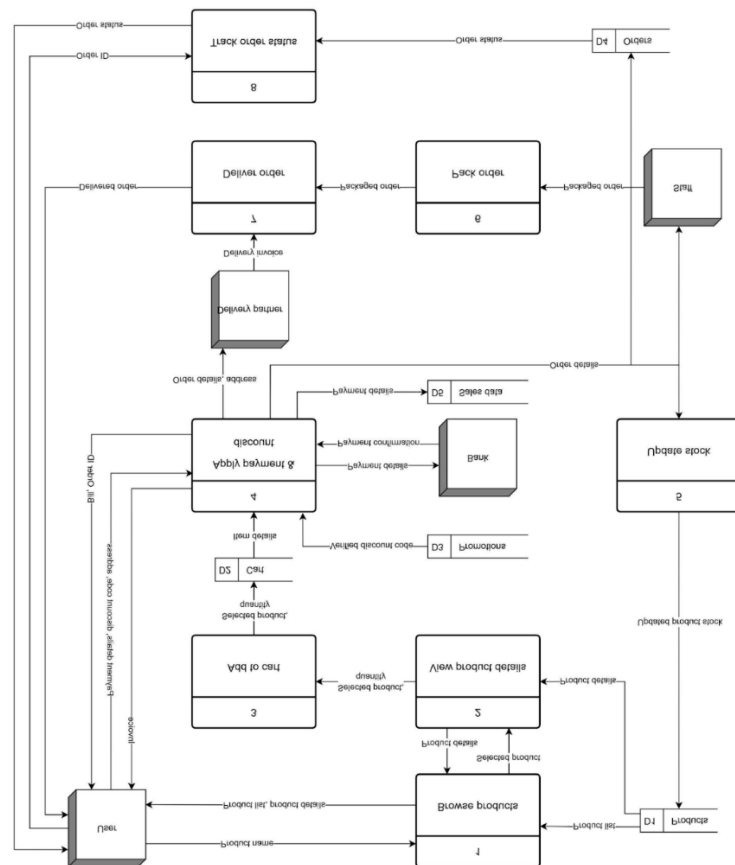


Figure 2. 10: Cart and order DFD.

For Cart and order, this DFD describes the sequence of operations a user follows from product selection to order delivery, as well as how data flows between the system's components during this process. The process begins when the user browses available products, which are retrieved from the Products data store. Upon selecting a specific product, the user can view its detailed information, such as specifications, images, and price. If interested, the user proceeds to add the product to their cart, including the selected quantity. This action updates the Cart data store with relevant item details.

When the user decides to place an order, they proceed to the payment stage, where they may also apply a promotional discount code. The system verifies this code against the Promotions data store and, if valid, updates the total cost accordingly. The user then submits payment details, which are sent to the bank for processing. Upon receiving confirmation, the system finalizes the order and saves it in the Orders data store. At this point, stock levels are updated in the Products database, and a new entry is added to the Sales Data collection to record the transaction.

Order fulfillment begins with the packaging step, which is handled by internal staff. Once the order is packaged, it is handed over to a delivery partner, who receives both the package and the associated delivery invoice. The delivery process concludes when the order is received by the user. Throughout this fulfillment process, users can track their order status, which is continuously updated based on information from the Orders data store.

c. Change user information

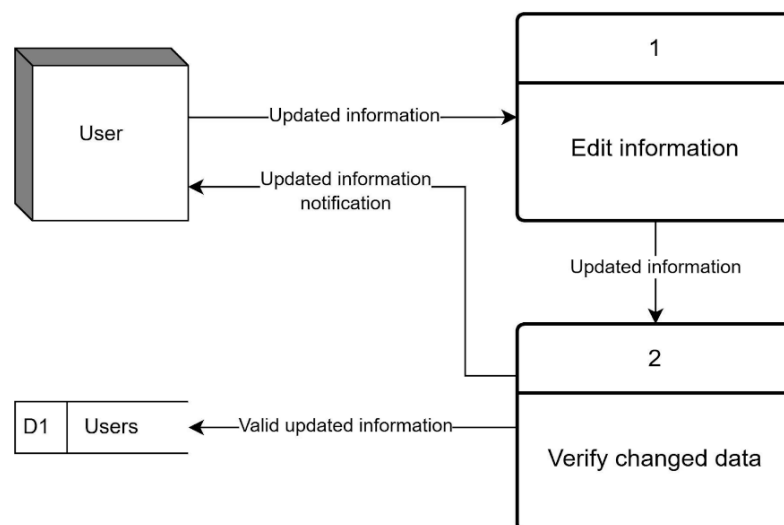


Figure 2. 11: Change user information DFD.

The DFD for changing user information outlines the process through which users update their personal profile details within the Collab Eyewear mobile application. The process begins when a user submits new or modified information, such as name, phone number, or default address, through the system interface. This data is directed to the "Edit Information" process, which prepares the changes for validation.

Next, the updated information is passed to the "Verify Changed Data" process. This step ensures that the submitted values are valid, properly formatted, and do not conflict with existing system rules (e.g., unique usernames or valid email formats). Once validated, the system stores the updated record in the Users data store, overwriting the old information with the new one. The user is then notified of the successful update.

d. Dynamic promotion

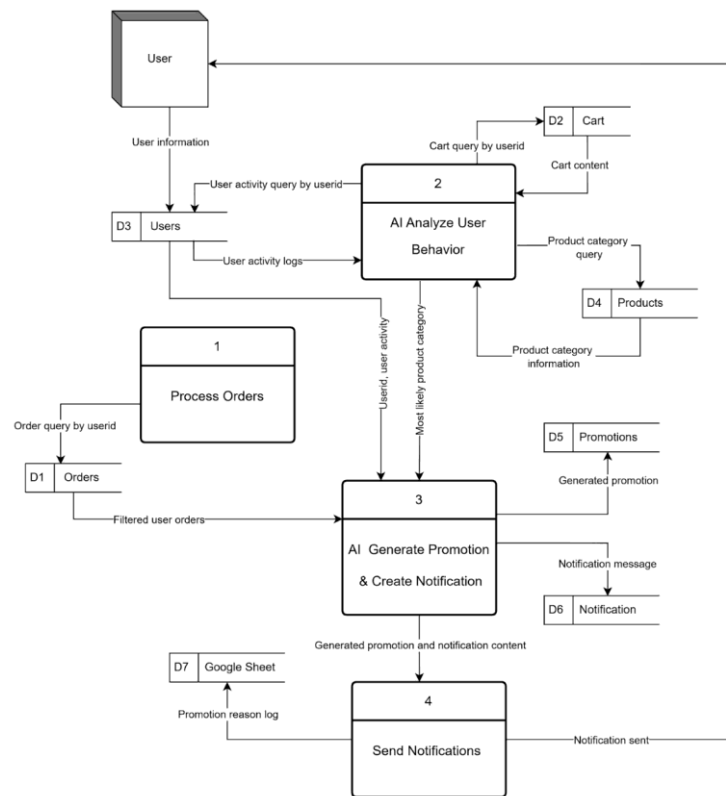


Figure 2. 12: Dynamic Promo DFD.

This DFD represents the personalized marketing mechanism within the Collab Eyewear system, which uses AI to analyze user behavior and generate targeted promotions. The process begins by querying order history from the Orders data store and retrieving the user's cart content and activity logs from the Cart and Users collections. This data is passed to the "AI Analyze User Behavior" module, which

evaluates past interactions, such as viewed products and items left in the cart, to determine the product category most relevant to the user's interest.

Based on this analysis, the system proceeds to the “AI Generate Promotion & Create Notification” process, where a suitable promotional offer is created. This promotion is tailored to the user’s most likely interests and is informed by current stock and promotional rules pulled from the Products and Promotions data stores. Simultaneously, a notification message is prepared and written into the Notifications data store for delivery.

Additionally, the promotion generation process logs its reasoning, such as behavioral patterns or purchase gaps, to a Google Sheet for administrative review and future optimization. Finally, the “Send Notifications” process delivers the personalized promotion directly to the user via the app, ensuring timely and contextually relevant engagement.

e. Admin

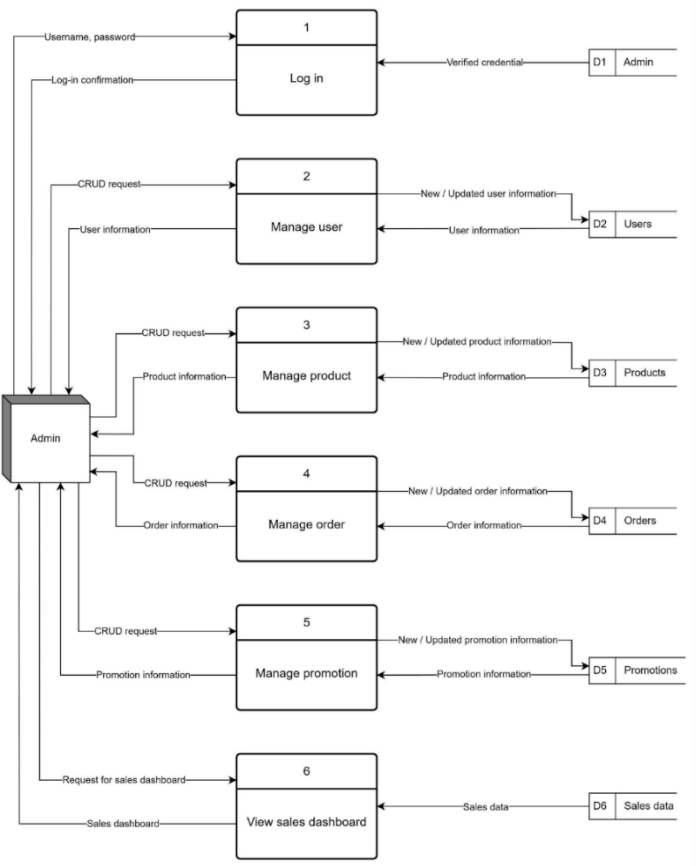


Figure 2. 13: Admin DFD.

The Admin DFD details the core administrative functions within the Collab Eyewear system, covering login authentication and the full suite of CRUD (Create,

Read, Update, Delete) operations for users, products, orders, promotions, and analytics. The process begins when the admin logs into the system using a username and password. These credentials are validated against the Admin data store, and upon successful verification, access is granted to the management dashboard.

Once authenticated, the admin can perform various management tasks. Managing user accounts involves creating new users, updating their details, or deleting accounts, all of which interact with the Users data store. Product management allows the admin to update product information, including pricing, images, and category assignments, which are written to the Products data store. Similarly, the admin can manage customer orders, tracking statuses or making adjustments, through the Orders data store.

Promotional campaigns can also be created, edited, or deactivated by the admin, with relevant data stored in the Promotions collection. Finally, the admin can request and view aggregated sales performance through a visual sales dashboard. This dashboard pulls real-time statistics and analytics from the Sales Data store to support decision-making and performance tracking.

2.4. Sitemap

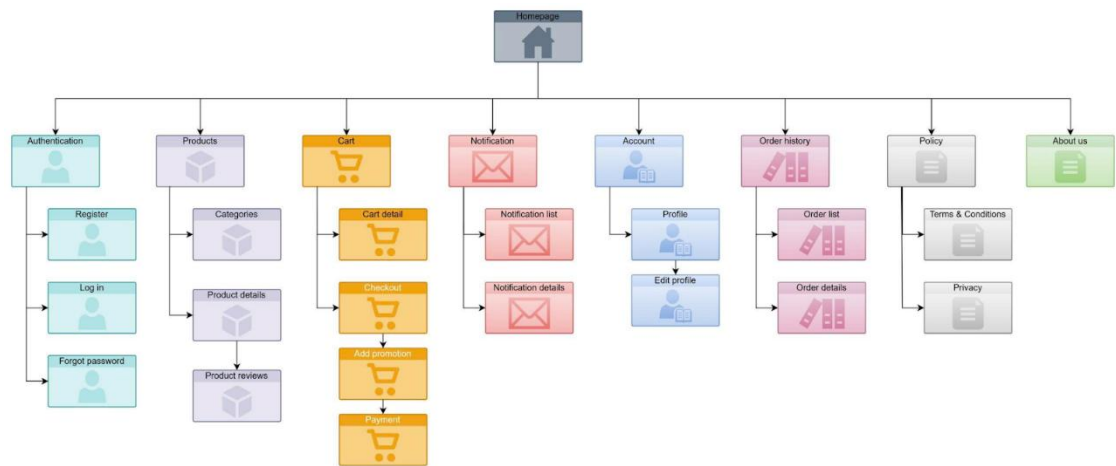


Figure 2. 14: Sitemap.

This sitemap outlines the overall structure and navigation hierarchy, showcasing how users interact with the system through various functional modules. At the top level, the homepage serves as the central entry point from which users can access key areas of the application, including Authentication, Products, Cart, Notification, Account, Order History, Policy, and About Us.

The Authentication section includes pages for registration, login, and password recovery, enabling secure access to the platform. The Products section offers category browsing and detailed product views, each of which leads to corresponding product reviews. Users can proceed to the Cart section, where they manage cart details, initiate checkout, apply promotional codes, and complete payments.

The Notification area provides access to a list of received messages, including promotional or transactional updates, along with detailed views for each notification. In the Account section, users can access their profile information and edit personal details such as contact information or shipping addresses. The Order History section allows users to view a list of previous orders and drill down into the details of each transaction.

To ensure transparency and compliance, the Policy section includes legal pages such as Terms & Conditions and Privacy Policy. Lastly, the About Us section offers information about the Collab brand and its mission.

2.5. Mockup

2.5.1. Logo and favicon design



Figure 2. 15: Image of Collab Logo

The Collab Eyewear logo reflects creativity, collaboration, and modern style. Its dynamic “COLLAB” typography suggests movement and innovation, while the small glasses around “AB” highlight the product focus. The deep green background with beige text adds elegance and warmth. More than a logo, it represents the brand’s mission: “Your Vision, Our Collaboration” - now brought to life in the design of our new mobile app.

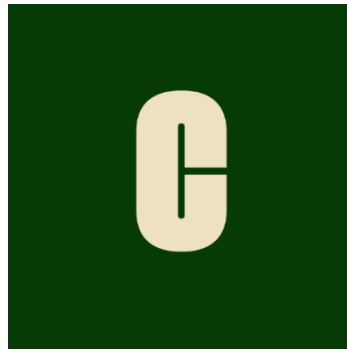


Figure 2. 16: Image of Collab Favicon

- Font: Montserrat and Playfair Display are two commonly used fonts in web design, each bringing a distinct style and visual character to a website.
- Color:

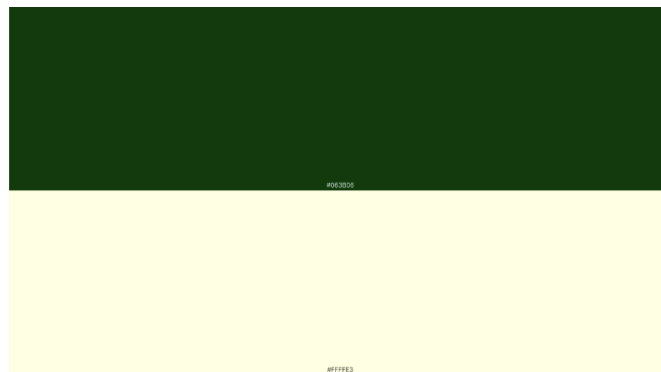


Figure 2. 17: Color palette.

2.5.2. Client side

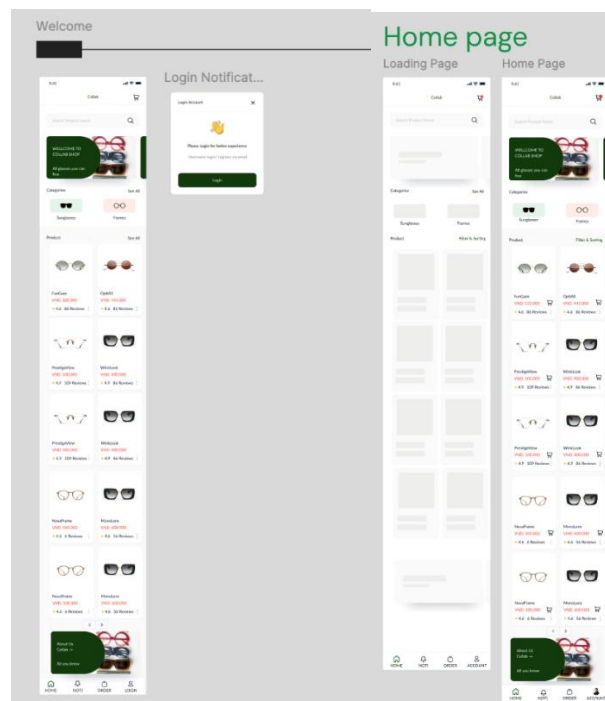


Figure 2. 18: Home Page (Login & None Login).

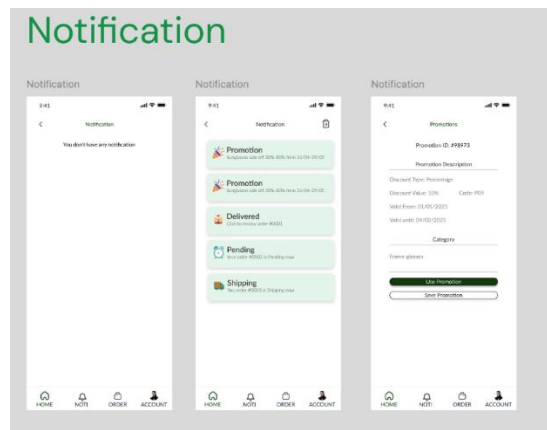


Figure 2. 19: Notification.

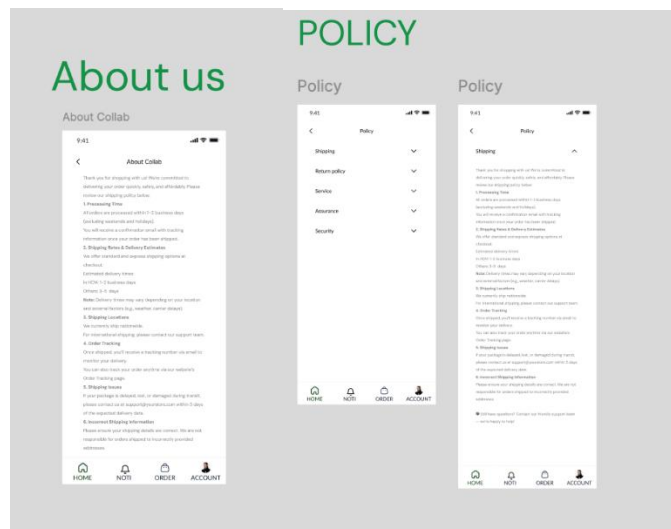


Figure 2. 20: About Us & Policy.

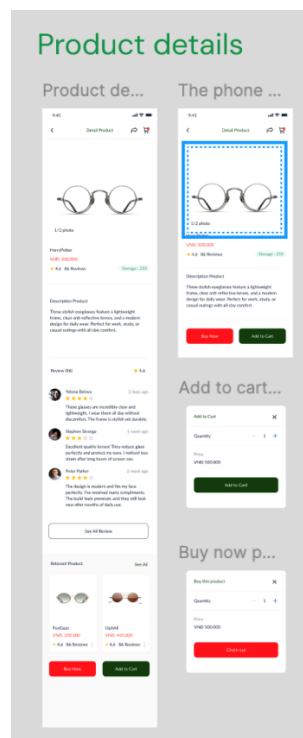


Figure 2. 21: Details Page and Component.

Product detail page and its components include: product image, product description, customer reviews, related products, and a purchase button. The screen also includes a popup to confirm either buying now or adding the product to the cart.

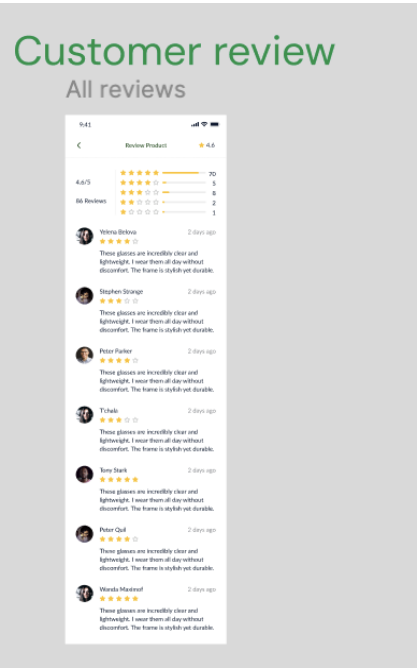


Figure 2. 22: Customer Reviews.

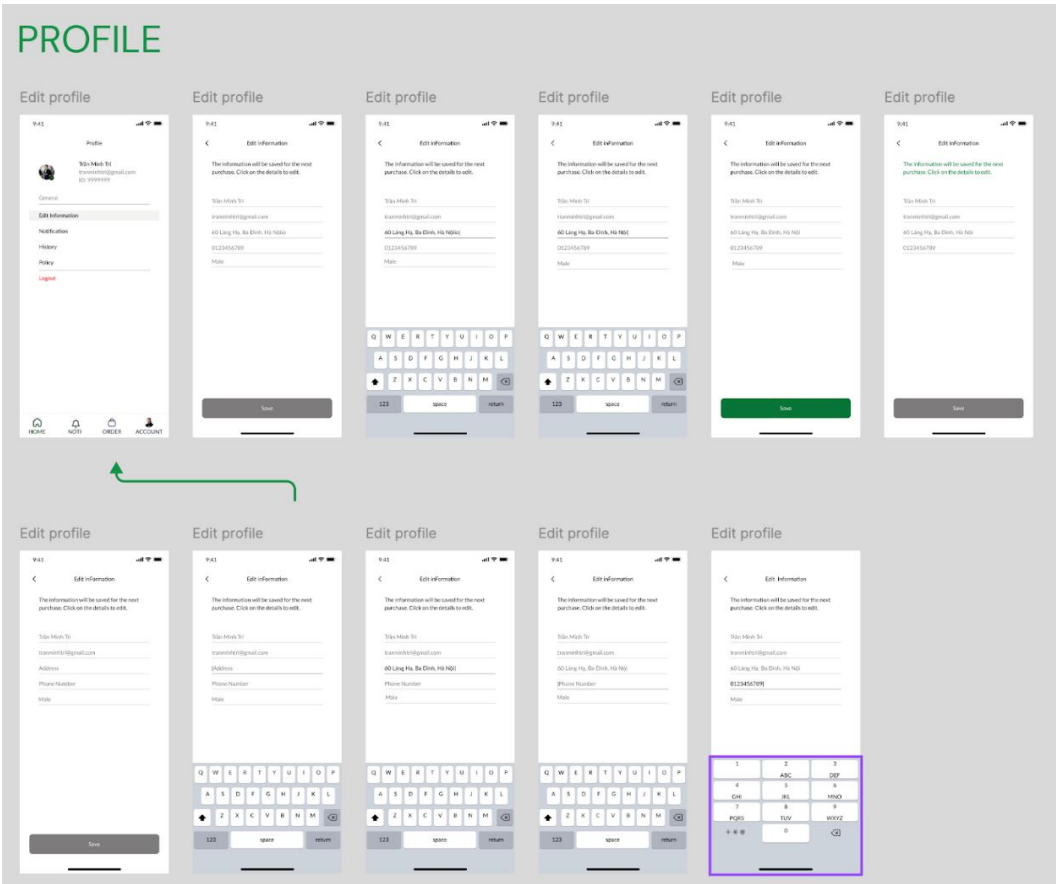


Figure 2. 23: Customer profile & Edit Information.

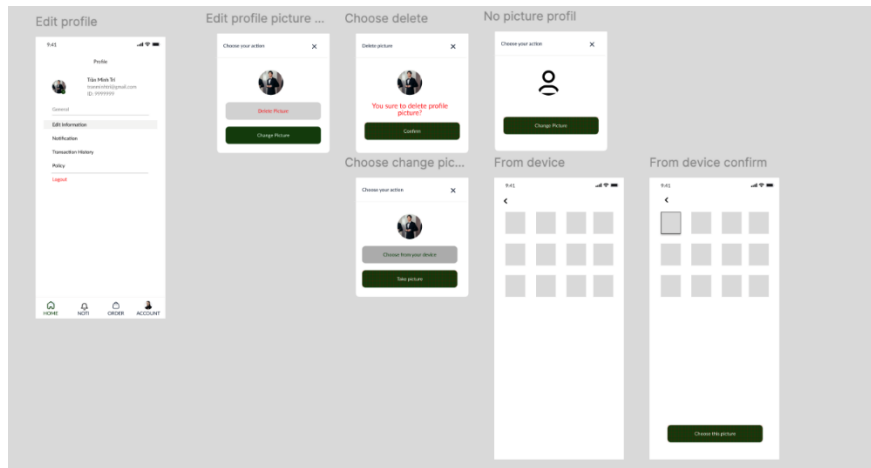


Figure 2. 24: Edit Profile picture.

Customers can either select an image from their device or take a photo using their phone's camera.

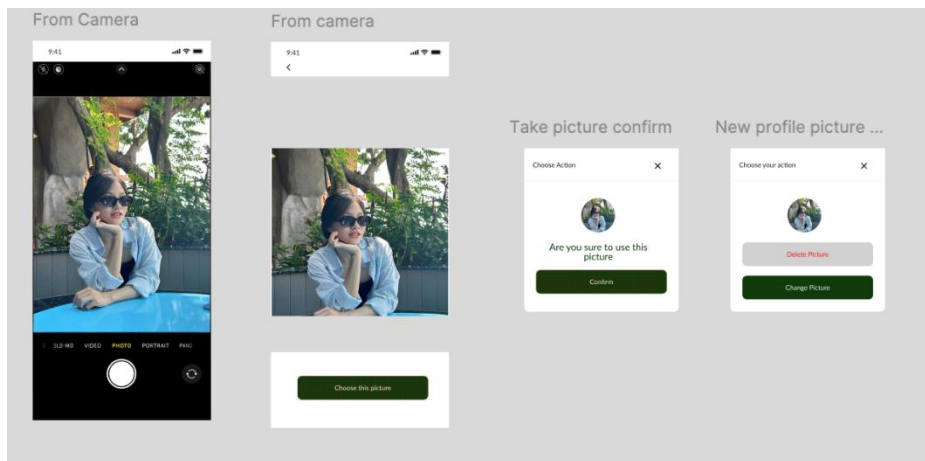


Figure 2. 25: Customer choose to take picture on the phone.

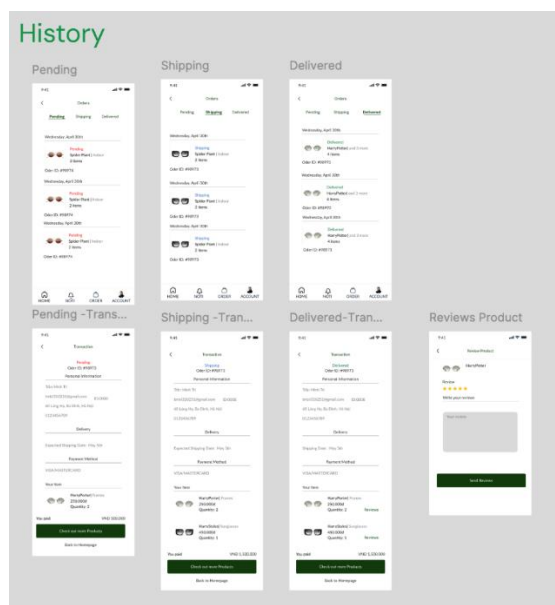


Figure 2. 26: History.

Each customer will have their order history saved, with three statuses: Pending, Shipping, and Delivered. When clicking on completed orders, they can review the products included.

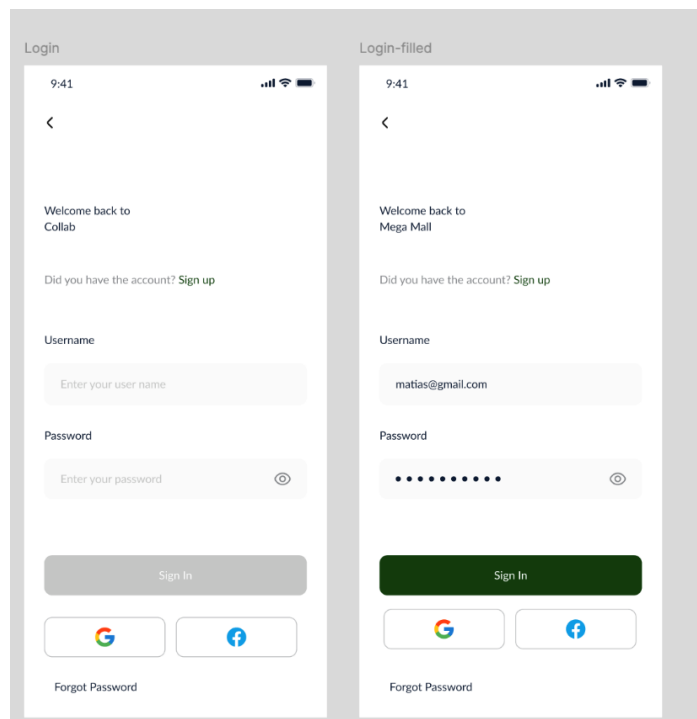


Figure 2. 27: Login.

User can log in via email, facebook, or sign-up.

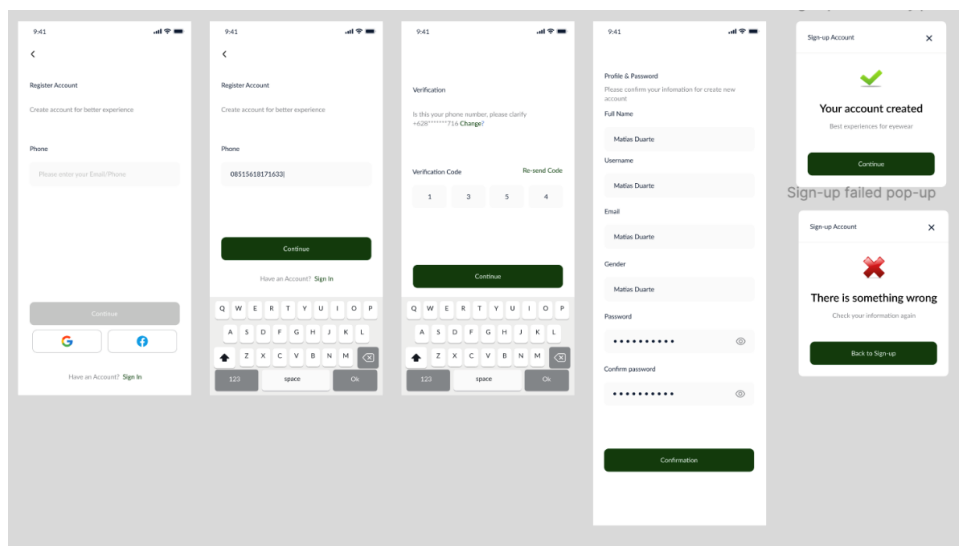


Figure 2. 28: Sign-up new Account.

If the registration is successful, a success message will be displayed. If any information is missing or duplicates an existing account, the message "There is something wrong" will appear.

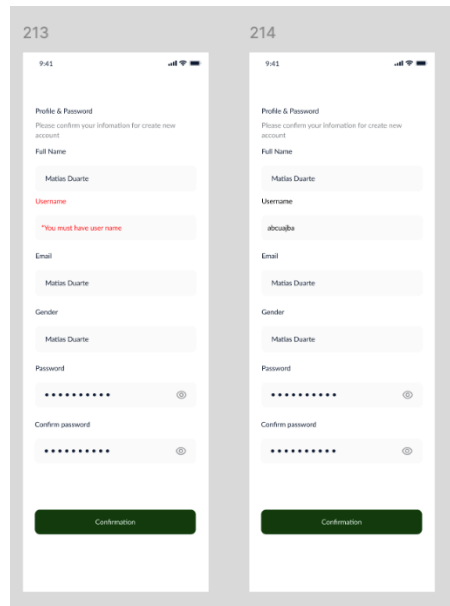


Figure 2. 29: Missing variable example.

If the information is incorrect or missing, the fields will be highlighted in red. They will return to normal once the input is complete and correct.

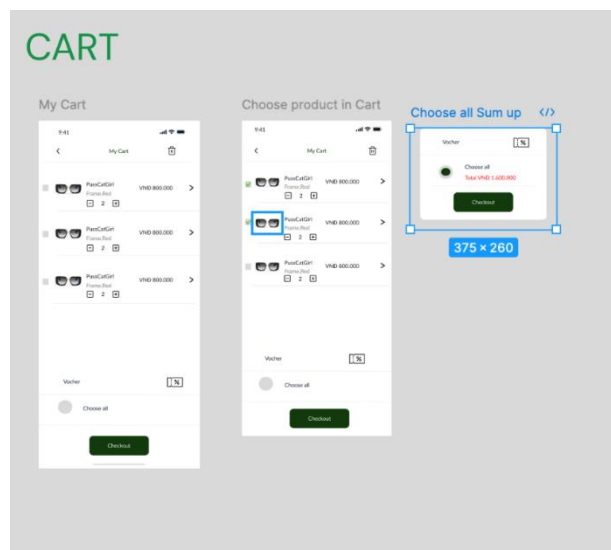


Figure 2. 30: Cart and Sum-up Pop-up.

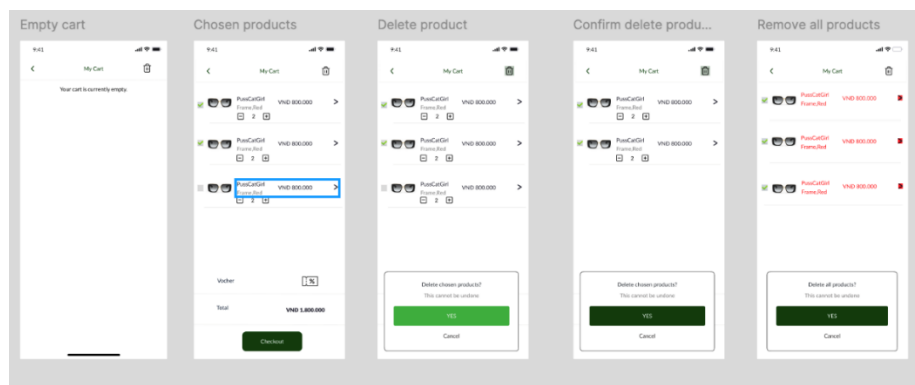


Figure 2. 31: Cart processing.

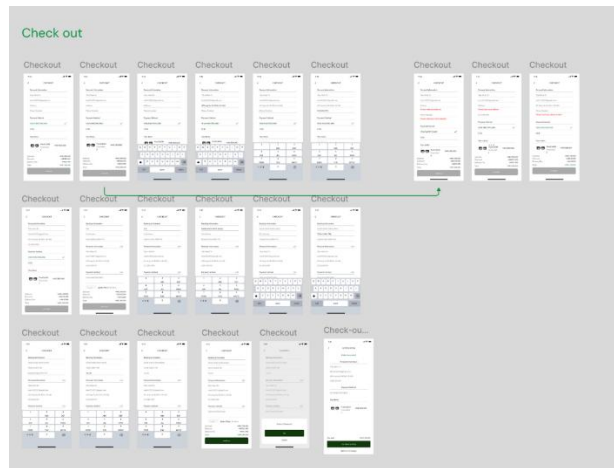


Figure 2. 32: Check out process.

Customers will check out from the cart. At this step, they can select the quantity of products to purchase, edit information, or choose the required address. Once the checkout is completed, a confirmation screen will appear showing the order has been placed successfully.

2.5.3. Admin side

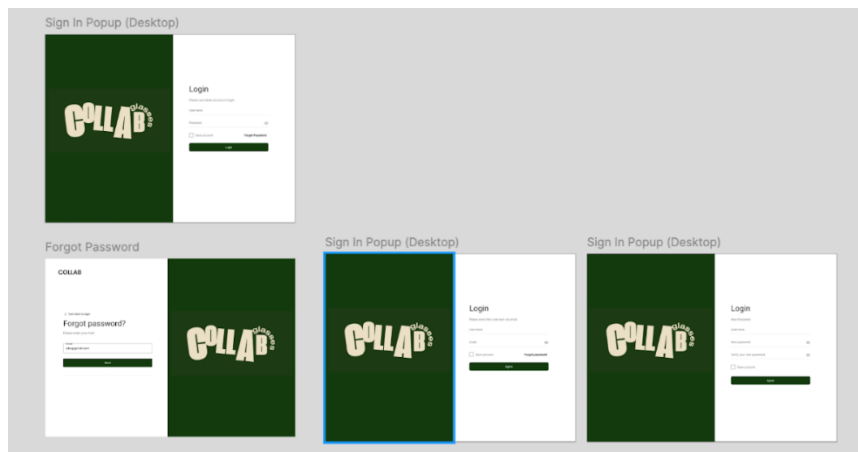


Figure 2. 33: Admin login design.

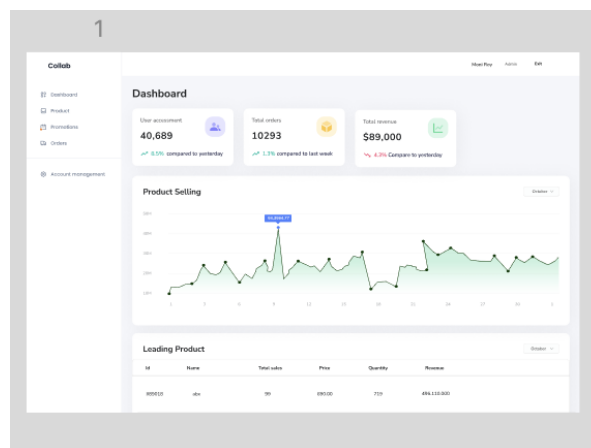


Figure 2. 34: Dashboard.



Figure 2. 35: Account Administration.

Both user accounts and admin accounts are managed. Admin accounts can also be added directly from this screen.

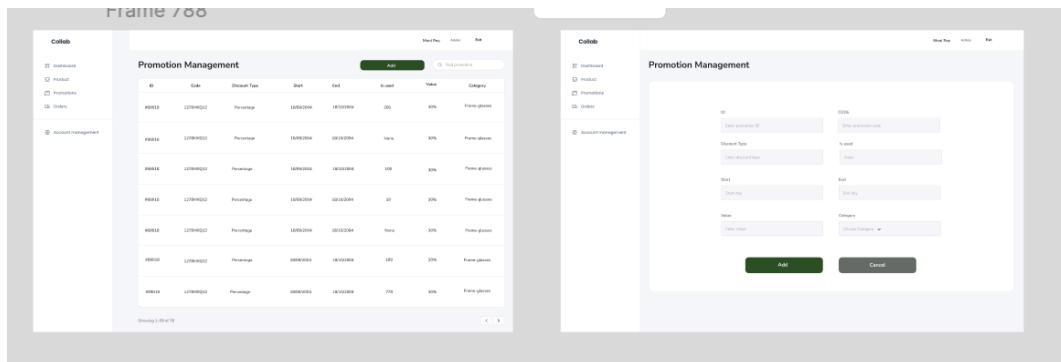


Figure 2. 36: Promotion Administration.

Containing the recent available promotion and admin can add new promotion in this page.

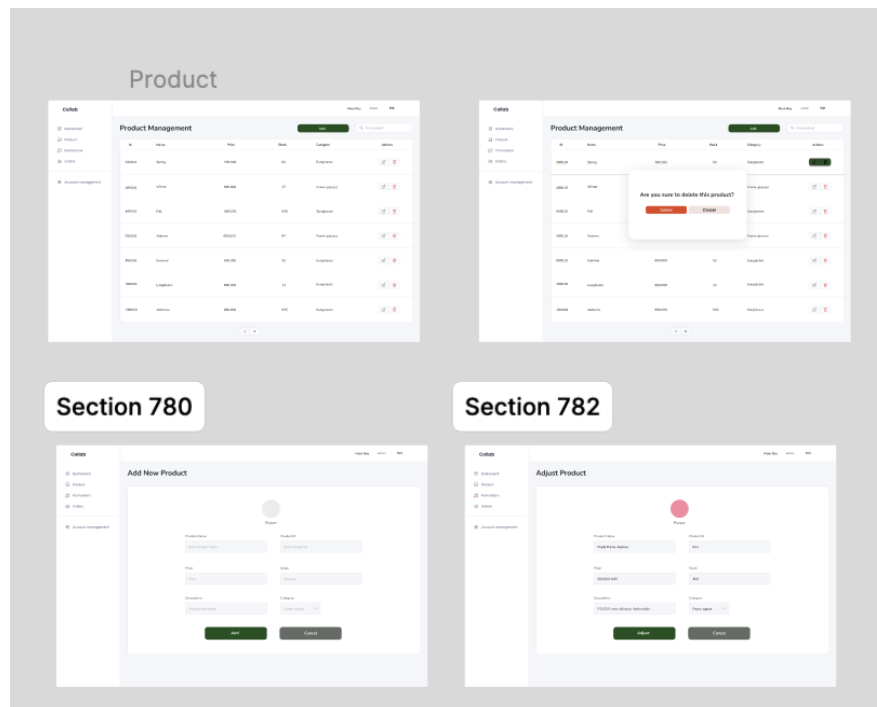


Figure 2. 37: Product Administration.

Manage the product quantity and allow adding new products directly within this section.

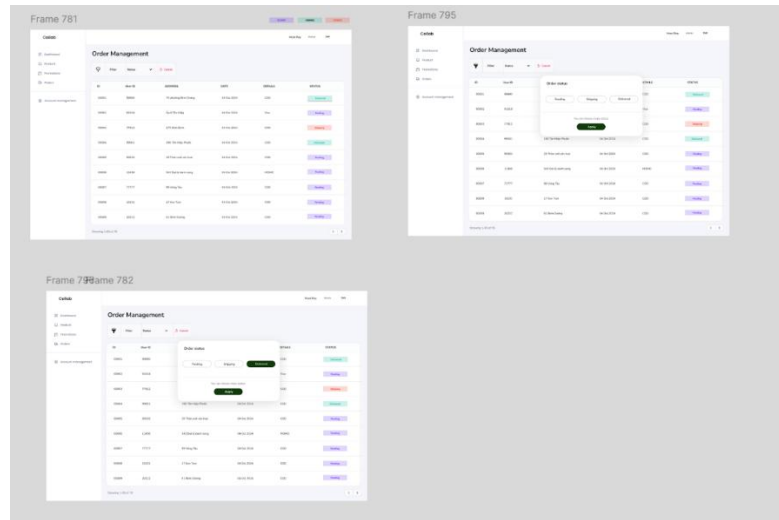


Figure 2. 38: Order Management.

Manage all orders, with specific order types displayed based on their status, and include a filter function to sort orders by status.

2.6. n8n AI agent

To enable real-time personalized promotions, the system implements a cloud-native automation workflow using n8n, integrated with a hybrid AI Agent that combines rule-based logic and large language model (LLM) reasoning. The architecture of this flow is illustrated in Figure 2.39.

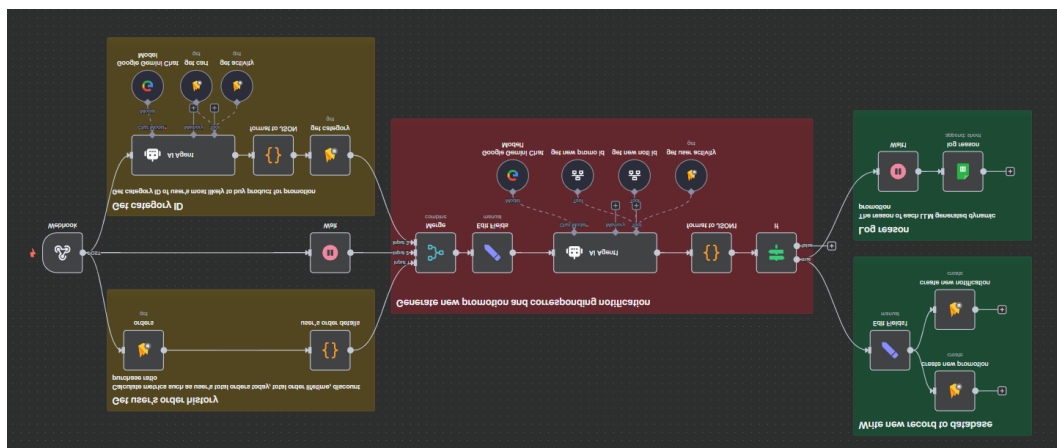


Figure 2. 39: n8n dynamic promotion workflow.

The dynamic promotion workflow is initiated by a webhook trigger in n8n, which activates two parallel data retrieval processes from Firebase. The first process fetches the user's order history to obtain variables such as total_orders_today, total_orders_lifetime, and discount_purchase_ratio. Simultaneously, the second process

gathers user activity data - such as recent product views and cart interactions - and predicts a likely categoryid for promotion using an auxiliary AI Agent.

These inputs are then merged and passed into the central component titled *“Generate new promotion and corresponding notification”*, which contains the system’s core logic. At the heart of this module is the AI Agent, powered by a large language model (Google Gemini) and operating based on a structured rule-based prompt. This hybrid approach allows the Agent to follow predefined logic for user classification and promotion eligibility, while still leveraging the LLM’s flexibility for context interpretation and message generation.

If eligible, the Agent classifies the user (new or returning), determines the appropriate discount type (fixed or percentage), and selects a value within a defined range. The value is chosen based on behavioral signals - such as high or low intent - so that users showing strong engagement are offered lower discounts, while disengaged users receive stronger incentives. The output includes two objects: a structured promotion and a corresponding notification message tailored to the user type and discount logic.

Once generated, n8n assigns unique IDs, timestamps (validfrom and randomized validuntil), and saves both records to Firebase Realtime Database. Additionally, the system logs the reasoning behind each decision to a Google Sheet, ensuring transparency and traceability for all LLM-generated promotions.

CHAPTER III. DEVELOPMENT APPLICATION

This chapter presents the system development process for the Collab Eyewear e-commerce platform, including setting up a Firebase database, creating sample data, and building functional modules for both the user and administrator sides. The developed features fully meet the application requirements such as displaying product catalogs, managing shopping carts, applying promotions, logging in/registering, as well as administrative functions such as product management, promotions, and order tracking.

3.1. Create sample data for Firebase Realtime

Detailed structure and examples of the stored data can be found in the Appendix E.

3.2. Develop system modules

3.2.1. *Client side*

a. Login

Login requires users to enter their previously registered username and password. After entering the information, the system will check. If the information is correct, the user will be redirected to the Homepage to start the shopping process. Conversely, if the username or password is incorrect, an error message will appear, asking the user to check and re-enter the correct information to continue.

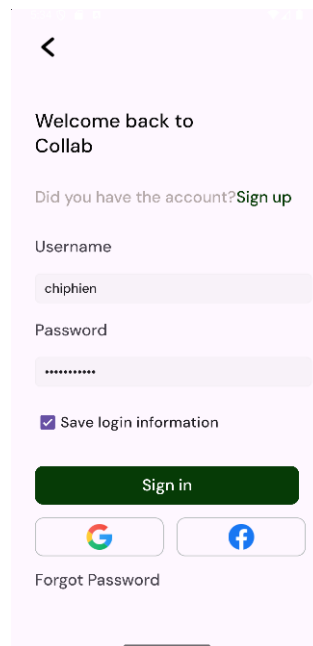


Figure 3. 1: Login page.

If the user forgets the password, the application provides a password reset feature right on the login interface. The user just needs to select the "Forgot password?" option to start the process. The system will then ask for the phone number registered with the account. A verification will be sent to allow the user to reset the new password. Once verified, the user will be redirected to the update page and enter the new password.

Figure 3. 2: Forgot password.

b. Register

If the user does not have an account, they will select the "Sign Up" button right on the login page. This action will redirect the user to the registration page. Here, the first step in the registration process requires the user to enter his/her phone number. The system will proceed to verify the phone number. If the verification code is entered correctly and valid, the user will be redirected to the next step of filling in personal information.

Figure 3. 3: Register account.

After completing the phone number verification step, users will be redirected to the registration information interface. Here, they need to fully enter personal information including: full name, username, email, gender, password and confirm password. These information fields are required to ensure completeness and accuracy during the account creation process.

Figure 3. 4: Create account.

After the user completes filling in the information and clicks the button to confirm creating an account, the system will display a pop-up message with the content "Sign-up Account". This message is to confirm that the registration process went smoothly.

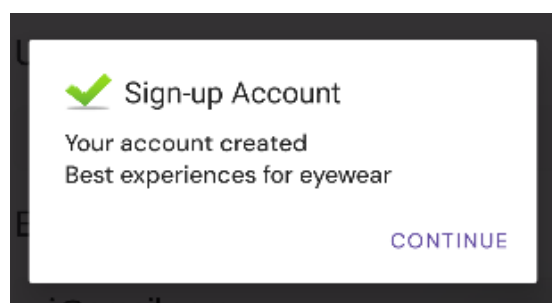


Figure 3. 5: Popup confirms successful account registration.

c. Welcome page

The Welcome Page is the first interface that users see when accessing the application without logging in. This page has the same layout and design as the main Homepage, to ensure consistency in user experience. However, the functionality of the

page is limited, only allowing users to access static content such as: viewing terms, policies, about Collab, product catalog, product details or searching products.

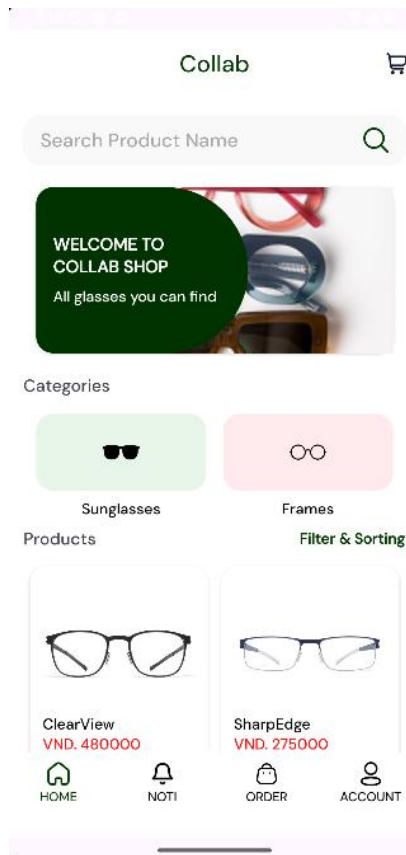


Figure 3. 6: Welcome page.

For functions that require interaction such as adding products to the cart, writing or viewing product reviews, and accessing order history, users are required to log in. When users attempt to perform these actions without logging in, the system will display a warning asking them to log in to continue using the function.

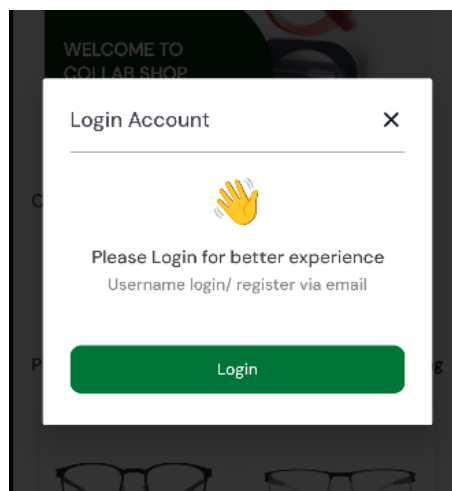


Figure 3. 7: Pop-up login.

d. Homepage

The Homepage is the main interface that logged-in users will access after accessing the application. In terms of design, the Homepage has the same layout as the Welcome Page to ensure consistency in user experience. The main components of the Homepage include:

- Search bar: Allows users to enter keywords to search for products by name, type or characteristics. The search bar is placed in an easy-to-see position, making it easy for users to access and use.

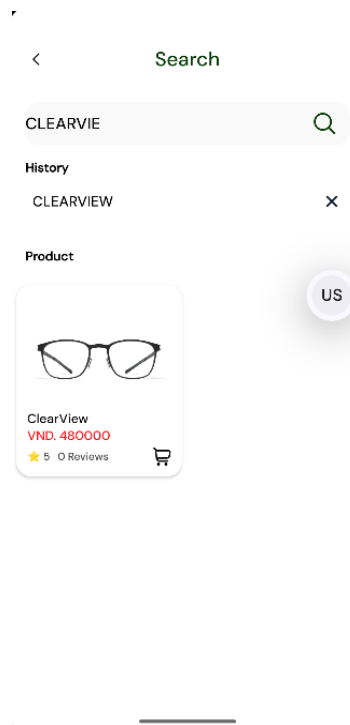


Figure 3. 8: Search bar.

- Product category: Products are divided into 2 main categories: Sunglasses and Frames. The classification helps users quickly access the product group that suits their needs without having to scroll through the entire list.



Figure 3. 9: Product category.

- Filter & Sorting: Users can filter products by many criteria such as price, classification and can sort results by price ascending/ descending, or product

name (A-Z/ Z-A). These features help improve the product browsing experience, helping users easily find the right product quickly and effectively.

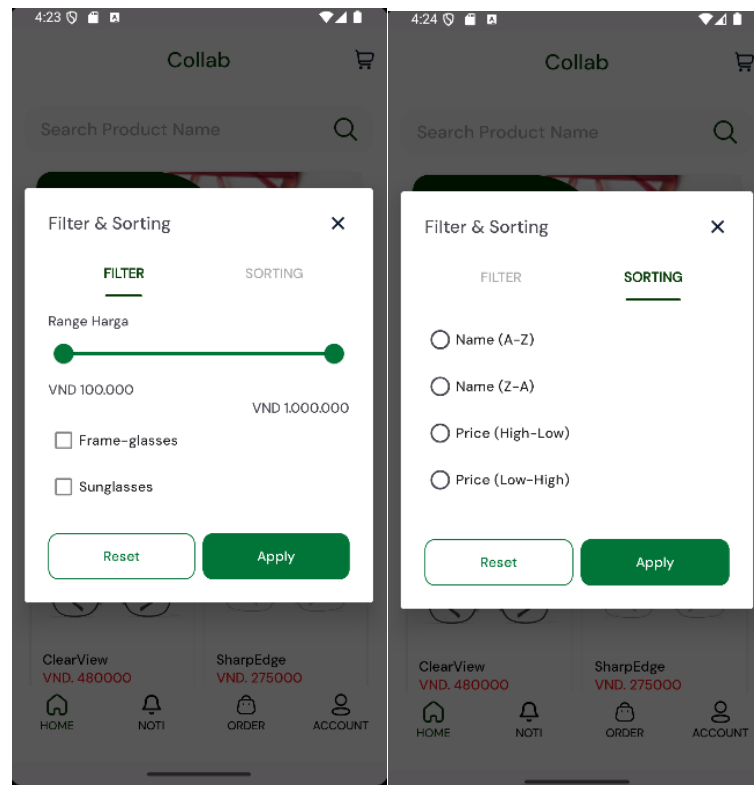


Figure 3. 10: Filter & Sorting.

e. Product details

The product detail page is where all the specific information of a product is displayed when the user clicks to select from the product list on the homepage. The interface of this page is designed intuitively, focusing on providing complete information so that users can easily make a purchase decision. The main content and functions of the page include: product image, product name and price, stock status, detailed description, reviews and comments and related products. In addition, there are 2 buttons such as:

- Buy Now: Take users directly to the payment process with the current product.
- Add to Cart: Add the product to the cart so that users can continue shopping and pay later.

The product detail page plays a key role in the shopping experience because it provides complete information as well as the necessary actions for users to quickly conduct transactions.

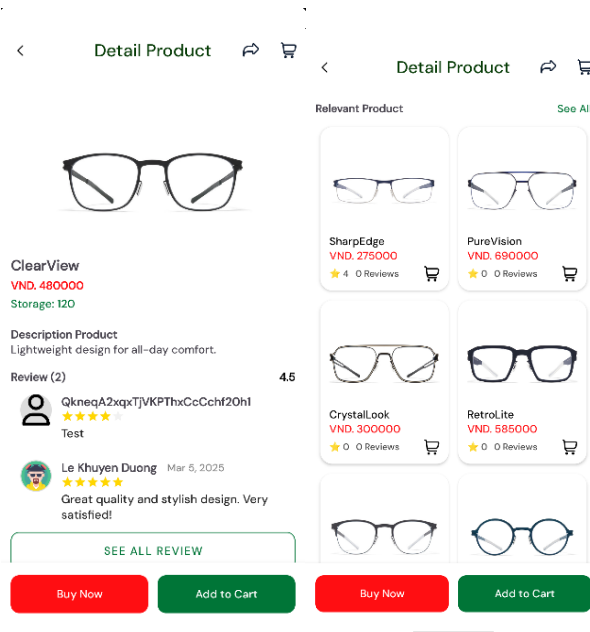


Figure 3. 11: Product details.

f. Profile

The Profile page is where all of the user's personal information is displayed after a successful login. Users can easily edit this information if there are any changes. In addition, they can also change their avatar by uploading a new image from their personal device. All updates are stored and synchronized on Firebase to ensure that the data is always accurate and secure.

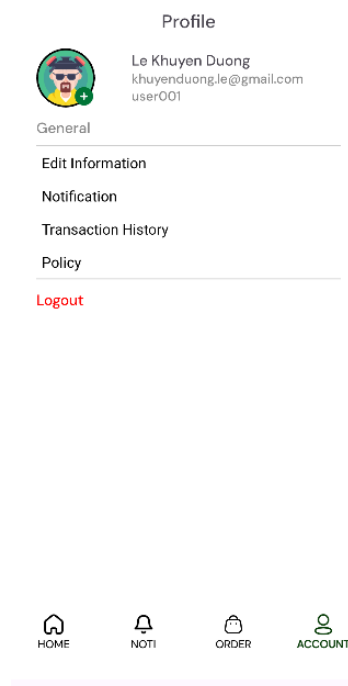


Figure 3. 12: Profile page.

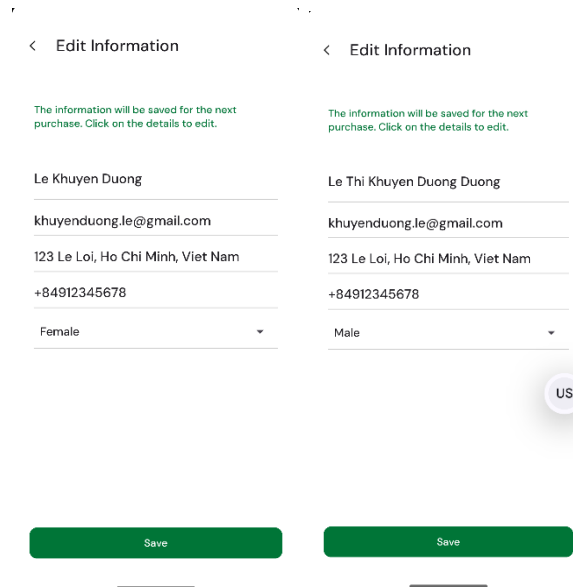


Figure 3. 13: Edit profile.

In addition, they can also change their avatar by uploading a new image from their personal device. All updates are stored and synchronized on Firebase to ensure that the data is always accurate and secure.

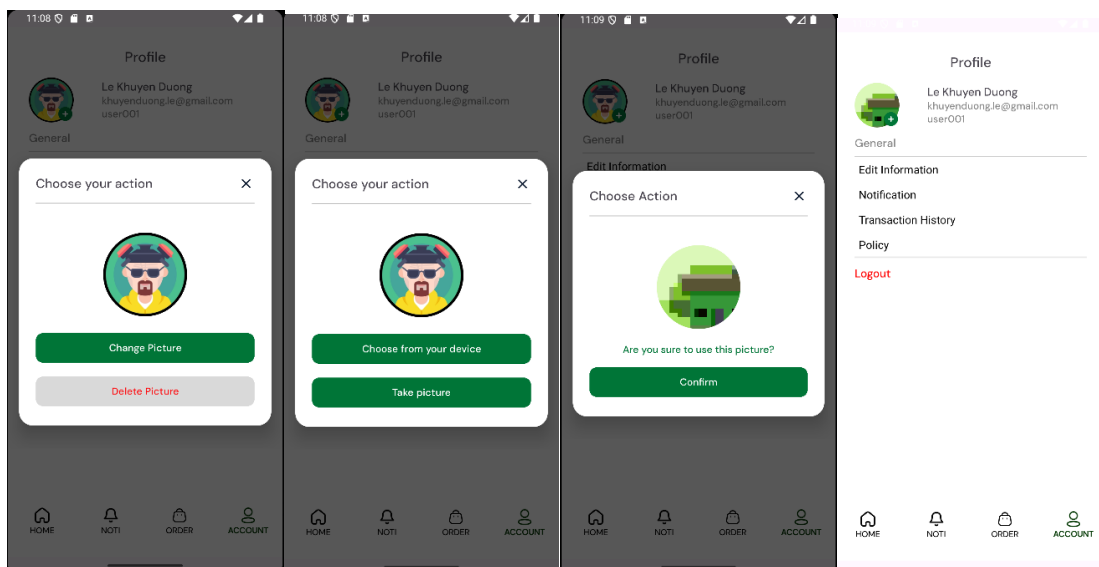


Figure 3. 14: Change avatar.

g. Notification

The Notification page is where all important notifications sent to users during the application are aggregated and displayed. The interface is divided into two main groups:

- Order status notifications: Including updates such as pending, shipping and delivered. These notifications help users track their shopping progress easily and promptly.
- Promotion notifications: Including two main types:

- **Public Promotion:** These are general discount or promotion programs that all users can use.
- **Specific Promotion:** These are personalized promotion codes, sent by the system to each user based on their purchase history or application usage behavior. These promotions help increase the level of personalization and attract users to return to purchase.

All notifications are updated in real time to ensure users do not miss important information.

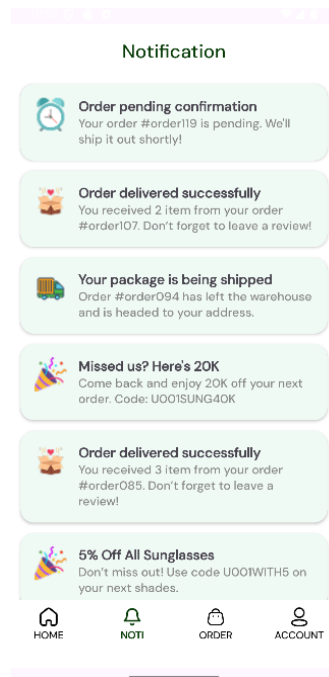


Figure 3. 15: Notification page.

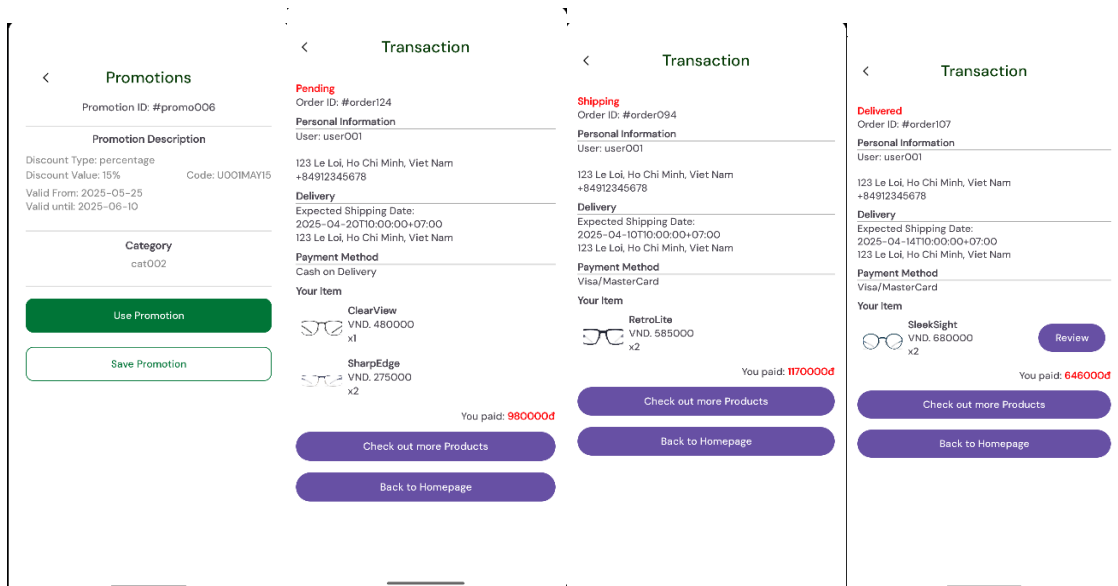


Figure 3. 16: Notification details.

h. Shopping cart page

The products will be automatically added to the cart after the user clicks the "Add to cart" button on the product detail page. Here, users can easily track the list of selected items, including images, product names, prices and totals corresponding to the quantity. The shopping cart interface supports the operation of increasing or decreasing the quantity of products according to actual needs, and also allows users to delete any products if they are no longer needed. In addition, users can also apply discount codes (vouchers) at this step to receive incentives, if any. After completing the adjustment, users can check the total cost of the order before moving to the payment step. The shopping cart page acts as a final checkpoint, helping users customize their purchase and ensure satisfaction before completing the transaction.

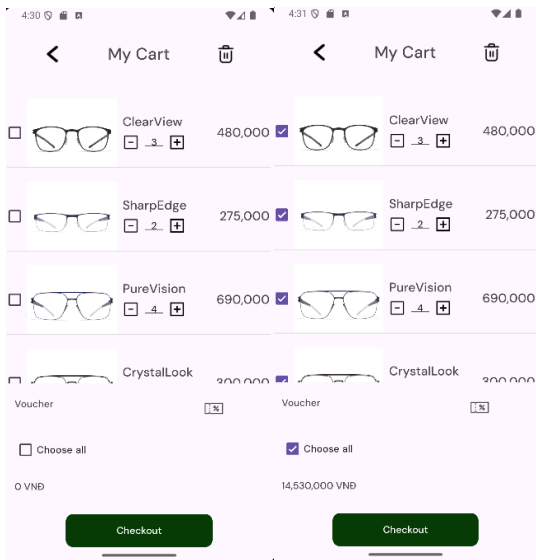


Figure 3. 17: Shopping cart page.

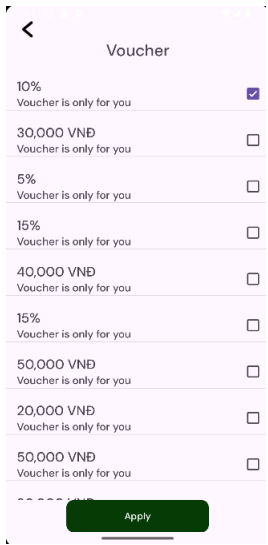


Figure 3. 18: List of vouchers.

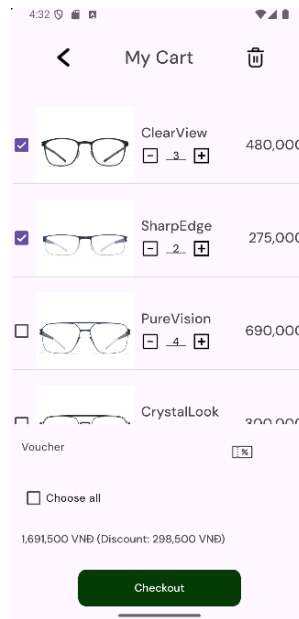


Figure 3. 19: Checking total cost, applying voucher before checkout.

i. Checkout

The checkout page is the final step in the shopping process. After checking the shopping cart and total cost, users are redirected to this page to complete the transaction. If the user has logged in before, the system will automatically fill in personal information from the saved personal profile, saving time and reducing errors. At this step, the system also provides many different payment methods for users to choose from, such as cash on delivery (COD), Visa/MasterCard. All order information such as products, quantity, price, shipping cost and total payment are clearly displayed, ensuring that users can easily check and confirm before completing the ordering process.

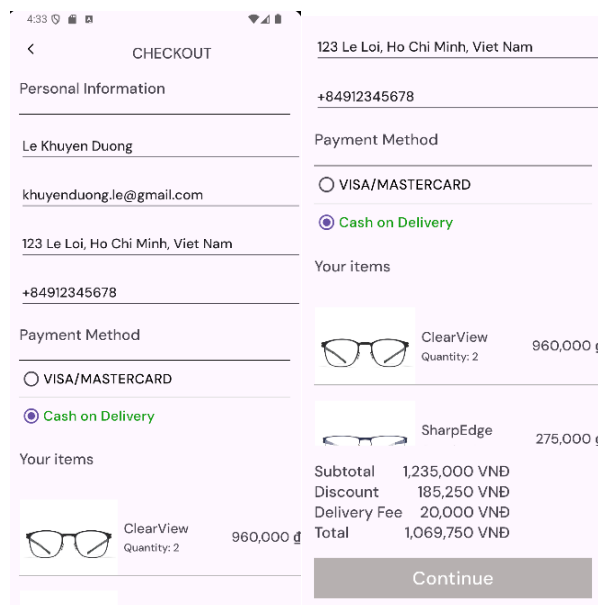


Figure 3. 20: Checkout.

If the user chooses COD (cash on delivery) as the payment method, after clicking confirm, the system will immediately process the order and transfer the user to the successful payment page, notifying that the order has been recorded and will be delivered as soon as possible. Conversely, if the user chooses to pay by VISA or MasterCard, the system will display an additional form to enter payment information.

The figure displays three sequential screenshots of a mobile application's checkout process. The first screenshot, titled 'CHECKOUT', shows a form for personal information (Le Khuyen Duong, khuyenduong.le@gmail.com, 123 Le Loi, Ho Chi Minh, Viet Nam, +84912345678) and payment method (VISA/MasterCard selected, Cash on Delivery unselected). It also lists items: ClearView (Quantity: 2, 960,000 đ) and SharpEdge (275,000 đ). The subtotal is 1,235,000 VNĐ, discount is 185,250 VNĐ, delivery fee is 20,000 VNĐ, and the total is 1,069,750 VNĐ. A 'Continue' button is at the bottom. The second screenshot shows the 'Banking Information' form with fields for card number (123456), name (Le khuyen Duong), and address (123 Le Loi, Ho Chi Minh, Viet Nam, +84912345678). The third screenshot shows the same checkout form as the first, but with the 'Continue' button highlighted.

Figure 3. 21: User chooses Visa/MasterCard payment method.

In addition, users can also edit personal information in the Edit Personal Information section.

The figure displays three sequential screenshots of a mobile application's checkout process, showing edited personal information. The first screenshot, titled 'CHECKOUT', shows a form for personal information (Le Chi Phien, khuyenduong.le@gmail.com, 123 Le Loi, Ho Chi Minh, Viet Nam, +84912345678) and payment method (VISA/MasterCard selected, Cash on Delivery unselected). It also lists items: ClearView (Quantity: 2, 960,000 đ) and SharpEdge (275,000 đ). The subtotal is 1,235,000 VNĐ, discount is 185,250 VNĐ, delivery fee is 20,000 VNĐ, and the total is 1,069,750 VNĐ. A 'Continue' button is at the bottom. The second screenshot shows the 'Banking Information' form with fields for card number (123456), name (Le khuyen Duong), and address (123 Le Loi, Ho Chi Minh, Viet Nam, +84912345678). The third screenshot shows the same checkout form as the first, but with the 'Continue' button highlighted.

Figure 3. 22: Edit personal information shipping.

After the user completes the payment steps such as selecting the delivery method, applying the discount code (if any), and selecting the payment method, the system will display a payment confirmation popup. This is the final step to ensure accuracy and avoid errors before proceeding to the official order.

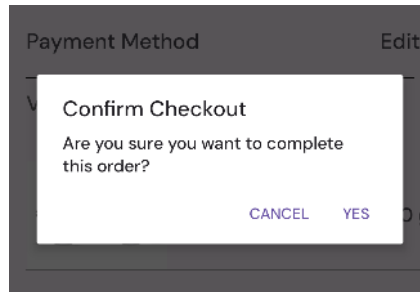


Figure 3. 23: Pop-up confirm checkout.

After the user clicks “Confirm” in the checkout popup step, the system will redirect to the “Order Successfully” page. Here, all information of the just-confirmed order will be clearly displayed. The page interface is friendly and easy to understand, helping users easily check the content of the order just placed. In addition, this page also provides two main navigation buttons: “Continue to Shop” to return to the product category and continue the shopping experience, and “Back to Homepage” to return to the main page of the application.

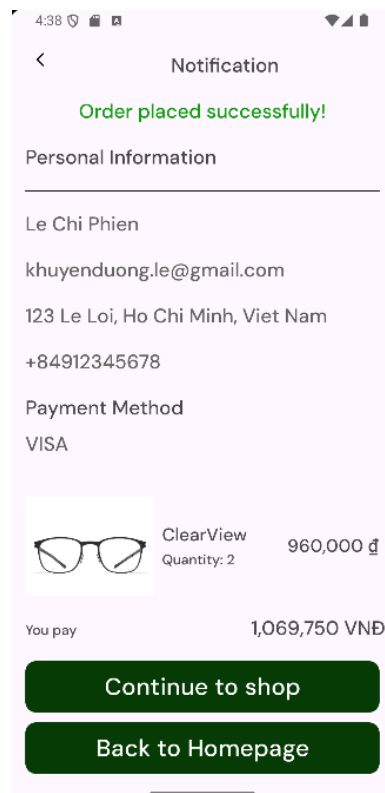


Figure 3. 24: Order successfully page.

j. Order history

The Order History page is where users can track their entire shopping history on the application clearly and intuitively. Here, orders are classified by status and all information for each order is displayed:

- Pending (Waiting for confirmation): Displays orders that users have just placed, waiting for system confirmation.
- Shipping (In progress): These are orders that have been confirmed and are in the process of being shipped to the buyer's address.
- Delivered (Delivered): Includes orders that have been successfully delivered to the user.

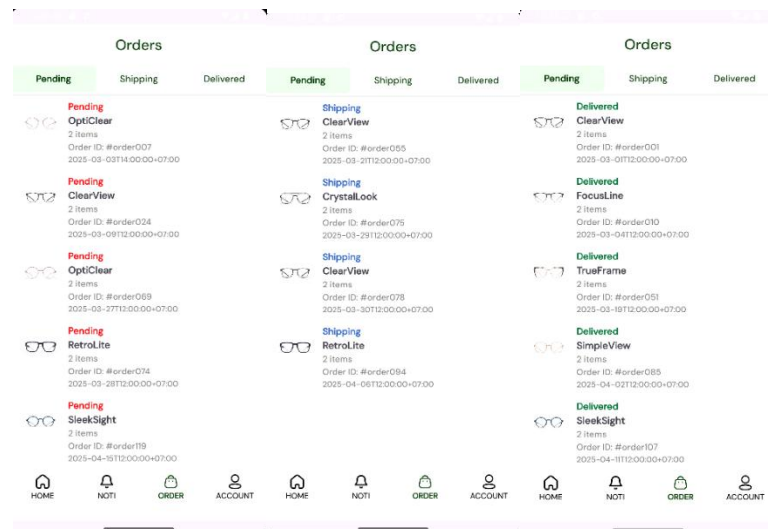


Figure 3. 25: Order history.

In particular, for orders with the Delivered status, the system will provide a link to the Product Review page. Here, users can rate and leave comments on the products they have received. This function is only open to products that have completed delivery, to ensure the authenticity of reviews from real buyers.

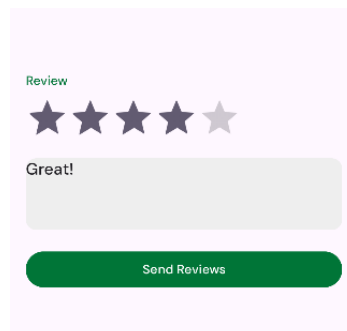


Figure 3. 26: Review product page.

k. Policy

The Policy page displays policies related to the use of Collab applications and services. The interface of this page is designed to be simple, clear and easy to read, helping users easily access important content.

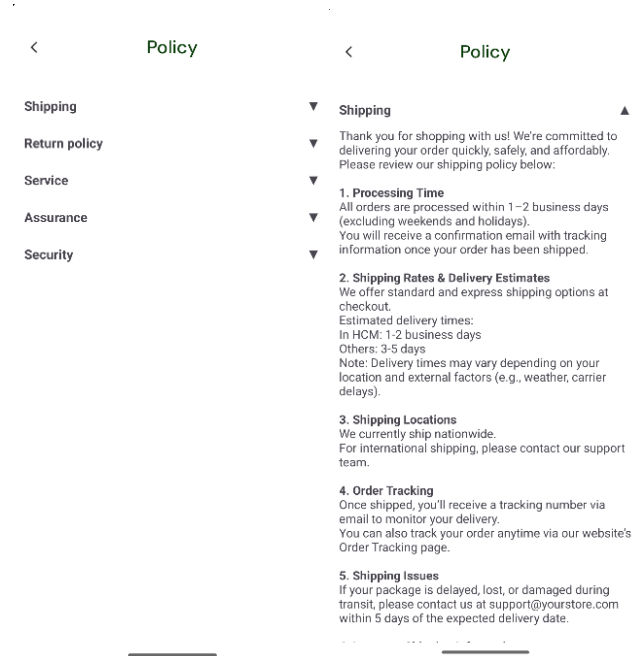


Figure 3. 27: Policy page.

3.2.2. Admin side

a. Login

The Admin Login page enables secure access for administrators via Firebase Authentication. They can log in with their credentials or reset a forgotten password using a verification code, ensuring uninterrupted access to the admin dashboard.

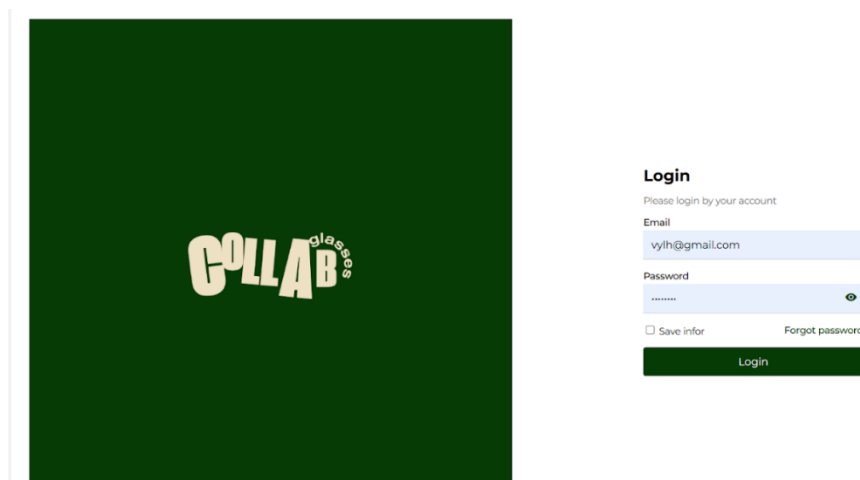


Figure 3. 28: Admin login page.

If admins forget their password, they have to fill in their email address to reset their password.

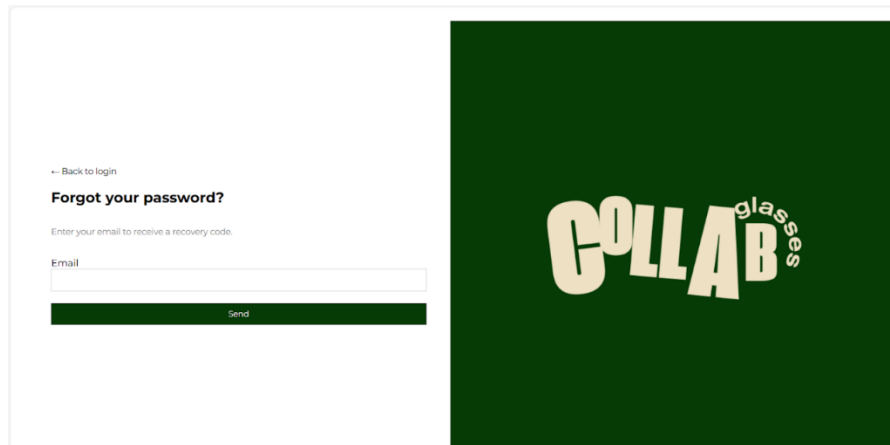
The image shows a web page for password recovery. On the left, there is a white form with a dark green header. The header contains a link "Back to login" and the title "Forgot your password?". Below the title, it says "Enter your email to receive a recovery code." There is an "Email" input field and a dark green "Send" button. On the right, there is a dark green square with the "COLLAB glasses" logo in white.

Figure 3. 29: Admin forgot password page.

They have to fill in a code that has been sent via their email.

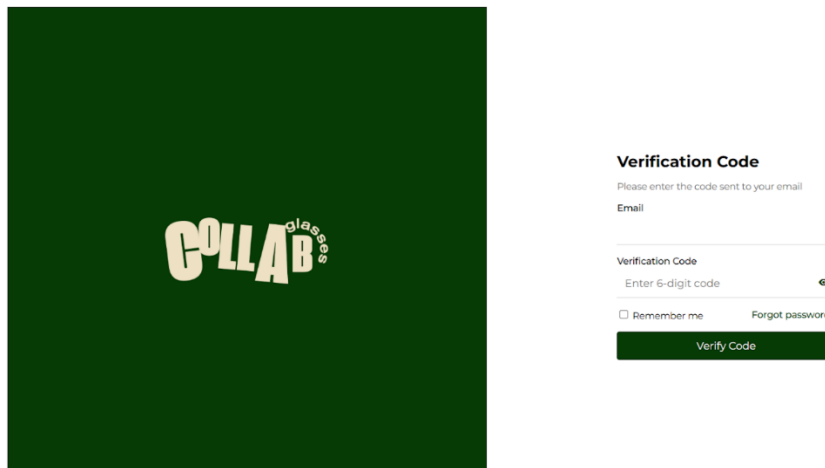
The image shows a web page for code verification. On the left, there is a dark green square with the "COLLAB glasses" logo in white. On the right, there is a white form with a dark green header. The header contains the title "Verification Code" and the instruction "Please enter the code sent to your email". Below the instruction, there is an "Email" input field. There is a "Verification Code" input field with the placeholder "Enter 6-digit code" and a toggle icon. Below the input field, there are two checkboxes: "Remember me" and "Forgot password?". At the bottom, there is a dark green "Verify Code" button.

Figure 3. 30: Admin confirm code page.

They can update a new password through filling in the bellowing form and link that has been sent to email.

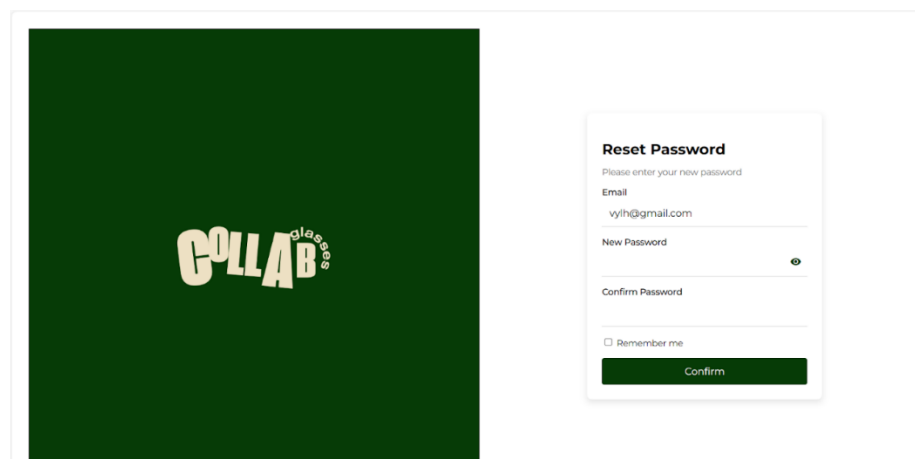
The image shows a web page for password reset. On the left, there is a dark green square with the "COLLAB glasses" logo in white. On the right, there is a white form with a dark green header. The header contains the title "Reset Password" and the instruction "Please enter your new password". Below the instruction, there is an "Email" input field with the placeholder "vylh@gmail.com". There is a "New Password" input field with a toggle icon. Below the input field, there is a "Confirm Password" input field. At the bottom, there is a checkbox "Remember me" and a dark green "Confirm" button.

Figure 3. 31: Admin reset password page.

b. Admin homepage

The Dashboard provides administrators with a comprehensive overview of sales performance, featuring critical metrics such as total revenue, order volume, and top-performing products. Additionally, it offers quick access to in-depth reports and analytics, empowering admins to track users application performance.

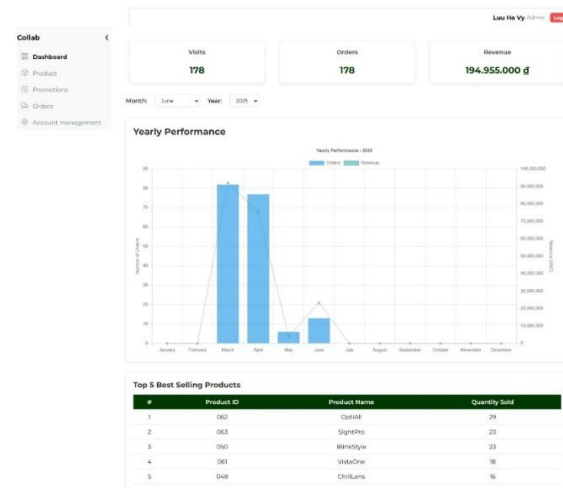


Figure 3. 32: Admin homepage with dashboard.

c. Product management

The Product Management interface enables administrators to perform full CRUD operations on product listings. This includes entering comprehensive product specifications, modifying existing entries, and archiving discontinued items from the digital catalog.

Product Management

Add Product

Search product...

ID	Name	Price	Stock	Category	Actions
#001	ClearView	₫480,000	120	Frame glasses	
#002	SharpEdge	₫275,000	120	Frame glasses	
#003	PureVision	₫690,000	120	Frame glasses	
#004	CrystalLook	₫300,000	120	Frame glasses	
#005	RetroLite	₫585,000	120	Frame glasses	
#006	FocusLine	₫410,000	120	Frame glasses	
#007	SleekSight	₫340,000	120	Frame glasses	

Page 1 / 10

Figure 3. 33: Product list.

Admin can update product information:

Product Name: ClearView

Product ID: 001

Price: 480000

Stock: 120

Description: Lightweight design for all-day comfort.

Category: Frame glasses

Buttons: Adjust, Cancel

Figure 3. 34: Editing product.

Admin can add new product:

Product Name: Enter product name

Product ID: 071

Price: 0

Stock: 0

Description: Enter product description

Category: -- Select Category --

Buttons: Adjust, Cancel

Figure 3. 35: Add new product.

Admin can delete product:

Product Management						Add Product	Search product...
ID	Name	Price	Stock	Category	Actions		
#001	ClearView	d480,000	120	Frame glasses	 		
#002	SharpEdge	d275,000	120	Frame glasses	 		
#003	PureV	Are you sure you want to delete product "ClearView"?			 		
#004	Crysta			glasses	 		
#005	RetroLite	d585,000	120	Frame glasses	 		
#006	FocusLine	d410,000	120	Frame glasses	 		
#007	SleekSight	d340,000	120	Frame glasses	 		

Figure 3. 36: Delete product.

d. Promotion management

The Promotion Management interface enables administrators to fully control discount campaigns through comprehensive CRUD functionality. Administrators can

configure promotion parameters including discount values, validity periods, and eligibility criteria, while maintaining real-time control over active customer offers.

Promotion Management

[Add Promotion](#)

ID	Code	Discount Type	Start	End	Used	Value	Category	Actions
promo001	U001THANKU10	percentage	15/03/2025	25/03/2025	Used	10	Frame glasses	Edit Delete
promo002	U001FRAME30K	fixed	25/03/2025	05/04/2025	Used	30000	Frame glasses	Edit Delete
promo003	U001WITH5	percentage	01/04/2025	10/04/2025	Unused	5	Sunglasses	Edit Delete
promo004	U001VIP15	percentage	25/03/2025	05/04/2025	Used	15	Frame glasses	Edit Delete
promo005	U001SUNG40K	fixed	05/04/2025	10/04/2025	Unused	40000	Sunglasses	Edit Delete
promo006	U001MAY15	percentage	25/05/2025	10/06/2025	Unused	15	Sunglasses	Edit Delete

Page 1 / 9

Figure 3. 37: Promotion list.

Admin can edit promotion information.

Promotion Management

ID:

CODE:

Discount Type:

Is used:

Start:

End:

Value:

Category:

[Update](#) [Cancel](#)

Figure 3. 38: Editing promotion information.

Admin can add new promotion:

Promotion Management

ID:

CODE:

Discount Type:

Is used:

Start:

End:

Value:

Category:

[Add](#) [Cancel](#)

Figure 3. 39: Add a new promotion.

Admin can delete promotion.

Promotion Management

[Add Promotion](#)

ID	Code	Discount Type	Start	End	Used	Value	Category	Actions
promo001	U001THANKU10	percentage	15/03/2025	25/03/2025	Used	10	Frame glasses	
promo002	U001FRZ	Are you sure you want to delete this promotion?			Used	30000	Frame glasses	
promo003	U001V				Unused	5	Sunglasses	
promo004	U001VIP15	percentage	25/03/2025	05/04/2025	Used	15	Frame glasses	
promo005	U001SUNG40K	fixed	05/04/2025	10/04/2025	Unused	40000	Sunglasses	
promo006	U001MAY15	percentage	25/05/2025	10/06/2025	Unused	15	Sunglasses	

Page 1 / 9

Figure 3. 40: Delete promotion.

e. Order management

The Order Management page enables administrators to monitor and handle customer orders efficiently. It provides access to comprehensive order details and allows admins to update the status of each order to indicate its current progress.

Order Management

[Filter](#) [Status](#) [Cancel](#)

Order ID	Customer	Delivery Address	Order Date	Payment Method	Status
order021	Le Duong Bao Lam	45 Nguyen Trai, Dong Nai, Viet Nam	Mar 8, 2025	Visa/MasterCard	Delivered
order022	Trinh Sang	35 Dan Toc, Vu Han, Trung Quoc	Mar 8, 2025	Visa/MasterCard	Shipping
order023	Ngoc Kem	22 Ly Thuong Kiet, Ho Chi Minh, Viet Nam	Mar 8, 2025	Cash on Delivery	Delivered
order024	Le Khuyen Duong	123 Le Loi, Ho Chi Minh, Viet Nam	Mar 9, 2025	Visa/MasterCard	Pending
order025	Vo Thi Chi	12 Vo Thi Sau, Vung Tau, Viet Nam	Mar 9, 2025	Cash on Delivery	Pending
order026	Unknown User	9 Tran Hung Dao, Can Tho, Viet Nam	Mar 9, 2025	Cash on Delivery	Delivered
order027	Ngo Mi Nga	100 Cach Mang Thang 8, Ha Noi, Viet Nam	Mar 10, 2025	Cash on Delivery	Delivered
order028	Vo Thi Chi	12 Vo Thi Sau, Vung Tau, Viet Nam	Mar 10, 2025	Visa/MasterCard	Delivered
order029	Trinh Tran Phuong Tuan	78 Hai Ba Trung, Ben Tre, Viet Nam	Mar 11, 2025	Visa/MasterCard	Pending
order030	Le Duong Bao Lam	45 Nguyen Trai, Dong Nai, Viet Nam	Mar 11, 2025	Cash on Delivery	Shipping

Page 3 / 18

Figure 3. 41: Order list.

Admin can filter order via order status:

Choose order status

☒ Pending ☒ Shipping ☒ Delivered

*Can choose many types of status

Apply

Figure 3. 42: Filter order based on status

f. Account management

The Account page enables administrators to oversee staff and admin accounts by viewing details such as staff ID, username, email. It also allows them to manage user permissions, ensuring each individual has appropriate access to perform their designated tasks on the site.

Account Management

Admin Accounts **Customer Accounts**

Add Admin

ID	Username	Email	Last Login	Actions
admin001	mienduong	vyth@gmail.com	5/17/25, 9:00 AM	 
admin002	camtu	ngannguyen230604@gmail.com	5/16/25, 3:42 PM	 
admin003	lieunhuyen	tiennnb@gmail.com	5/15/25, 9:18 PM	 
admin004	thuquyen	quyenmtt@gmail.com	5/14/25, 11:25 AM	 
admin005	duongtang	vyntt@gmail.com	5/13/25, 6:03 PM	 

Page 1 / 1

Figure 3. 43: Admin account list.

Adding a new admin account.

Add New Admin

Username

Password

Email

Add **Cancel**

Figure 3. 44: Add a new admin.

They can edit info and delete account.

ID	Username	Email	Last Login	Actions
admin001	mienduong	vyth@gmail.com	5/17/25, 9:00 AM	 

Figure 3. 45: Edit info admin.

Are you sure you want to delete "mienduong"?

Confirm **Cancel**

Figure 3. 46: Delete admin account.

CHAPTER IV. CONCLUSION

The project “E-commerce application for Collab eyewear brand” has been implemented by the team in a systematic and comprehensive manner, from requirements analysis, system design to building and realizing the complete product. The main goal of the project is to build an e-commerce platform that aims at a friendly and convenient user experience, while providing an effective management system for businesses. The system consists of two main parts: a mobile application for end users (client app) and an administration page for administrators (admin web).

On the user side, the team uses Android Studio to develop a mobile application running on the Android platform. This application allows users to perform main functions such as: register/login, browse product categories (sunglasses, frames), view product details, add products to cart, apply promotion codes, make payments, track order status and rate products after receiving the goods. The interface is designed to be friendly and easy to use, combining a notification and promotion system to help personalize the user experience.

On the administrator side, the system is built using Angular - a modern framework developed by Google, specializing in developing single-page web applications (SPA). The administrative interface allows the admin to fully perform functions such as: managing products, categories, promotions, tracking orders, managing user information and revenue statistics with intuitive charts.

Both systems use Firebase Realtime Database as a platform for storing and synchronizing data in real time. Firebase provides a user authentication solution (Firebase Authentication), stores data in JSON format and supports flexible expansion, which is very suitable for the dynamic characteristics of e-commerce systems.

In particular, the project also integrates a dynamic promotion feature using AI Agent combined with n8n - an open source tool that allows building automation processes. Thanks to that, the system can automatically send personalized promotions to each user based on product browsing behavior and purchase history, thereby improving marketing efficiency and order conversion rate.

Through the project implementation process, the team effectively applied the knowledge learned in mobile programming, web development, NoSQL database, API

integration, applied AI and system design. At the same time, the team also improved soft skills such as teamwork, systems thinking, progress management and problem solving.

Although the system has met the requirements, the team still sees a lot of potential for expansion and upgrade in the future, such as: developing an iOS version, integrating more diverse payment gateways, using machine learning to predict user needs, and supporting more effective inventory management.

In conclusion, the project not only helps the team better understand the process of developing a comprehensive e-commerce system, but also provides an opportunity to connect theory and practice, laying the foundation for practical applications in the field of Information Technology.

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APPENDIX A – DOCUMENT REFERENCES

See in

https://drive.google.com/drive/folders/1pZyU5cQ5O_rZWCTPSlArcLiCp8M8spFm?usp=sharing

APPENDIX B – GITHUB LINK FOR MOBILE APP

See in <https://github.com/luuhavy/collab-android-app.git>

APPENDIX C – GITHUB LINK FOR ADMIN WEB

See in <https://github.com/ngannguyen204/web-admin.git>

APPENDIX D – WEBSITE MOCKUP ON FIGMA

See in <https://www.figma.com/design/Lq7gkl7u7NzQfUDqZeYZDD/Collab-prototype--Copy-?node-id=123-2631>

APPENDIX E – FIREBASE REALTIME DATABASE

See in

<https://console.firebase.google.com/u/2/project/collabeyewears/database/collabeyewears-default-rtdb/data/~2F>

APPENDIX F – SCIENCE PAPER

See in

https://drive.google.com/drive/folders/1QZfNIeN7_pkR5SM1wWpTuVCVIZO5jkRh?usp=sharing

APPENDIX G – EVINDENCE CLIP

See in <https://drive.google.com/drive/folders/1txgIqCwBc3GCp-rX1HdEQTYPyxjjTGp?usp=sharing>

APPENDIX H – PRESENTATION FILE

See in https://drive.google.com/drive/folders/1gXRZTksYU22_g-53ePHVEnjgbYIIHVW?usp=sharing

APPENDIX I – DIAGRAMS

See in <https://drive.google.com/drive/folders/1iyvH31bGJ6VxZcXj1Pi9Dskl7TpgN8ij?usp=sharing>

APPENDIX K – EXCEL WORKING CONTRIBUTION

See in <https://docs.google.com/spreadsheets/d/1kTNZeuoElTojBmmY4gGENMVIMsKvYlQ7/edit?usp=sharing&ouid=115426758956083202940&rtpof=true&sd=true>